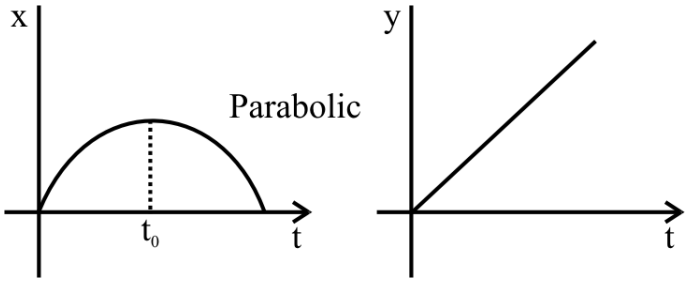
PHYSICS

1)

An object is moving in x-y plane it position-time graphs are given. Select the correct statement :-



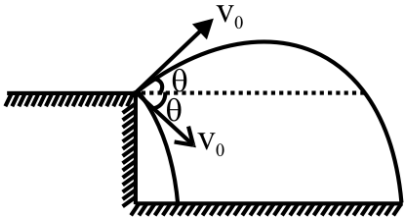
(1) motion of object is non-uniformly acceleration

(2) x-co-ordinate is continuously increasing

(3) speed is maximum at time t_0

(4) at time t_0 velocity and acceleration are perpendicular

2) Two balls are thrown with same speed v_0 from the top of a cliff. The angles of their initial velocities are θ above and below the horizontal as shown. How much farther along the ground does the top ball hit than the bottom ball :-



(1) $\frac{2v_0^2}{g}$

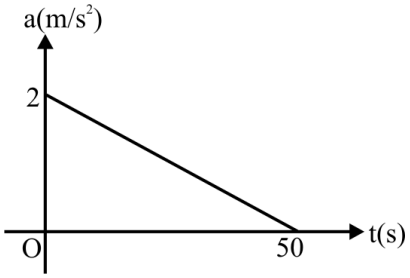
(2) $\frac{2v_0^2 \sin\theta}{g}$

(3) $\frac{2v_0^2 \sin\theta\cos\theta}{g}$

(4) $\frac{2v_0^2 \cos\theta}{g}$

3)

Referring to a-t diagram, find the velocity of the particle at t = 20 s. Assume that the particle was at the origin at t = 0 and was moving with a velocity of 10 m/s in –ve x-direction :-



(1) 20

(2) 22

(3) 18

(4) 15

4) Given $\vec{A} = 3\hat{i} + 4\hat{j}$ and $\vec{B} = 6\hat{i} + 8\hat{j}$. Which of the following statement is incorrect?

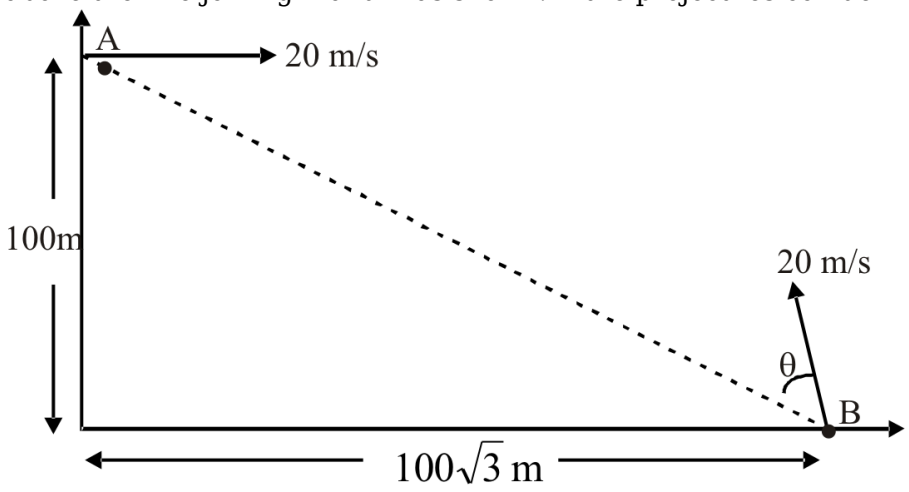
(1) $\vec{A} \cdot \vec{B} = 50$

(2) $\frac{|\vec{A}|}{|\vec{B}|} = \frac{1}{2}$

(3) $|\vec{A}| = 5$

(4) $\vec{A} \cdot \vec{B} = 48$

5) Two small balls A and B are launched in the same vertical plane simultaneously, with same speed of 20 m/s at t = 0. Ball A has an initial horizontal velocity and ball B has initial velocity at an angle θ above the line joining A and B as shown. If the projectiles collide in mid-air at time t :



(1) θ = 45°

(2) θ = 60°

(3) θ = 30°

(4) θ = 37°

6) An object is thrown horizontally from a tower H meter high with a velocity of $\sqrt{2gH}$ m/s. Its speed on striking the ground will be :

(1) $\sqrt{2gH}$

(2) $\sqrt{6gH}$

(3) $2\sqrt{gH}$

(4) $2\sqrt{2gH}$

7) A projectile is given initial velocity of $\hat{i} + 2\hat{j}$. The equation of the path is : (g = 10 m/s^2)

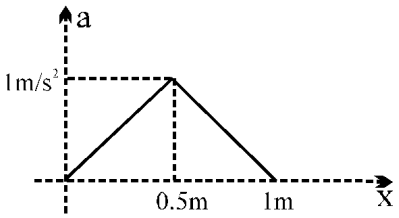
(1) $y = 2x - 5x^2$

(2) $y = x - 5x^2$

(3) $4y = 2x - 5x^2$

(4) $y = 2x - 25x^2$

8) A body initially at rest, starts moving along x-axis in such a way so that its acceleration vs displacement plot is as shown in figure. The maximum velocity of particle is :-



(1) 1 m/s

(2) 6 m/s

(3) 2 m/s

(4) none

9) The motion of a particle is defined by the position vector $\vec{r} = A(\cos t + t \sin t)\,\hat{i} + A\,(\sin t - t \cos t)\,\hat{j}$
Where t is expressed in seconds. Determine the value of t for which positions vectors and acceleration vector are perpendicular.

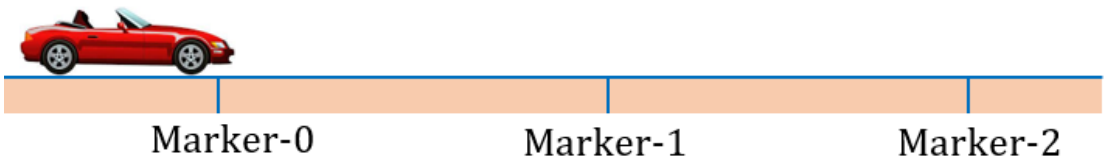
(1) 0

(2) 1

(3) 3

(4) 5

10) A car moving at 160 *km/h* when crosses the marker-0, driver applies brake and reduces its speed uniformly to 40 *km/h* at marker-2. The markers are spaced at equal distances along the road as shown below.
At which part of the track would the car have been traveling at instantaneous speed of 100 *km/h*?



(1) At marker-1

(2) Between marker-0 and marker-1

(3) Between marker-1 and marker-2

(4) More information required to decide.

11) The XI[th] class students of ALLEN designed a rocket. The rocket was launched from cycle stand of ALLEN straight up into the air. At $t = 0$, the rocket is at $y = 0$ with $V_y(t = 0) = 0$. The velocity of the rocket is given by : $V_y = (24t - 3t^2)$ *m/s* for $0 < t \leqslant t_b$ where t_b is the time at which fuel burns out completely. Vertically upward direction is taken as positive. ($g = 10$ *m/s*^{2})
The time taken by rocket to reach its maximum height is -

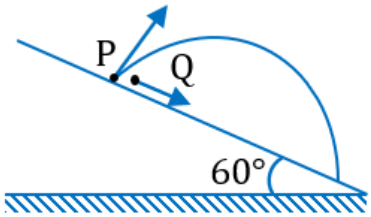
(1) 4 sec.

(2) 8 sec.

(3) 8.8 sec.

(4) 9.6 sec.

12) A particle *P* is projected from a point on the surface of smooth inclined plane (see figure). Simultaneously another particle *Q* is released on the smooth inclined plane from the same position. *P* and *Q* collide after $t = 4$ *s*. The speed of projection of *P* is :-



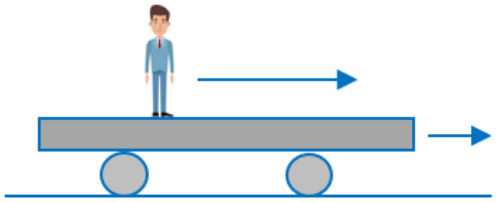
(1) 5 *m/s*

(2) 10 *m/s*

(3) 15 *m/s*

(4) 20 *m/s*

13) A trolley is moving horizontally with a constant velocity of *v* with respect to earth. A man starts running from one end of the trolley with a velocity 1.5 *v* with respect to trolley. After reaching the opposite end, the man returns back and continues running with a velocity of 1.5 *v* w.r.t. the trolley in the backward direction. If the length of the trolley is *L* then the displacement of the man with respect to earth during the process will be :-



(1) 2.5 *L*

(2) 1.5 *L*

(3) $\frac{5L}{3}$

(4) $\frac{4L}{3}$

14) A coin is released inside a lift at a height of 2 m from the floor of the lift. The height of the lift is 10 m. The lift is moving with an acceleration of 9 m/s^2 downwards. The time after which the coin will strike with the floor of lift is : (g = 10 m/s^2)

(1) 4 s

(2) 2 s

(3) $\frac{4}{\sqrt{21}}$ s

(4) $\frac{2}{\sqrt{11}}$ s

15) The trajectory of a projectile near the surface of the earth is given as $y = 2x - 9x^2$. If it were launched at an angle q_0 with speed n_0 then ($g = 10\ ms^{-2}$) : **[JEE-Main 2019]**

(1) $\theta_o = \cos^{-1}\left(\frac{1}{\sqrt{5}}\right)$ and $v_o = \frac{5}{3}ms^{-1}$

(2) $\theta_o = \sin^{-1}\left(\frac{1}{\sqrt{5}}\right)$ and $v_o = \frac{5}{3}ms^{-1}$

(3) $\theta_o = \sin^{-1}\left(\frac{2}{\sqrt{5}}\right)$ and $v_o = \frac{5}{3}ms^{-1}$

(4) $\theta_o = \cos^{-1}\left(\frac{2}{\sqrt{5}}\right)$ and $v_o = \frac{5}{3}ms^{-1}$

16) An airplane pilot wants to fly from city *A* to city *B* which is 1000 *km* due north of city *A*. The speed of the plane in still air is 500 *km*/hr. The pilot neglects the effect of the wind and directs his plane due north and 2 hours later find himself 300 *km* due north-east of city *B*. The wind velocity is

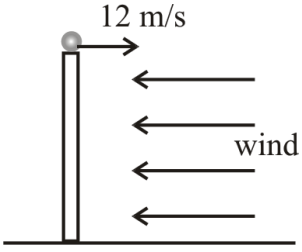
(1) 150*km*/*hr* at 45°*N* of *E*

(2) 106*km*/*hr* at 45°*N* of *E*

(3) 150 *km*/*hr* at 45°*N* of *W*

(4) 106 *km*/*hr* at 45°*N* of *W*

17) A ball is projected horizontally from the top of tower at 12 m/s as shown in figure wind is blowing as indicated (horizontally), which causes acceleration 6 m/s^2 horizontally. Displacement of the ball when it strikes the tower, (g = 10 m/s^2)



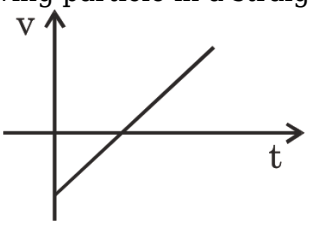
(1) 20m

(2) 60m

(3) 80m

(4) 40m

18) For a moving particle in a straight line the velocity time (v-t) curve is given as shown. Choose correct one :



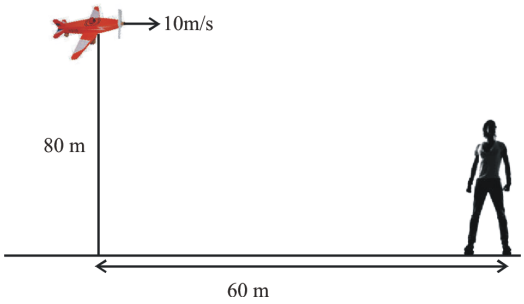
(1) The acceleration of particle firstly positive and then negative

(2) Particle's speed increasing continuously

(3) Initial position of particle was at left of origin.

(4) Particle changes its direction of motion once in a journey.

19) A toy plane is flying horizontally at 10 m/s at height 80 m above the ground toward a man on the ground as shown, when plane is 60m away horizontally from man, a ball is dropped by the plane. Find speed of man (assumed uniform) so that he is able to catch the ball just before it strikes the ground.



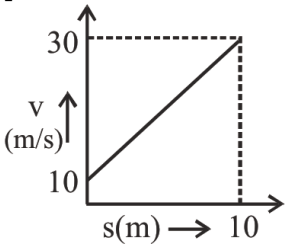
(1) 10 m/s

(2) 15 m/s

(3) 5 m/s

(4) 7.5 m/s

20) Velocity of a particle changes with position according to following curve. Acceleration of the particle at x = 1 m :-



(1) 24 m/s^2

(2) 2 m/s^2

(3) 20 m/s^2

(4) 3 m/s^2

21)

The relation between time t and distance x is $t = ax^2 + bx$, where a and b are constants. The acceleration is :

(1) $-2abv^2$

(2) $2bv^2$

(3) $-2av^3$

(4) $2av^3$

22) $\frac{d^2}{dx^2}\left(2x^2 \sin 30^\circ\right) = ...$

(1) 4x

(2) 2x

(3) 2

(4) 2x cos 30°

23) A stone is dropped from the top of a tall cliff and n seconds later another stone is thrown vertically downwards with a velocity u. Then the second stone overtakes the first, below the top of the cliff at a distance given by :

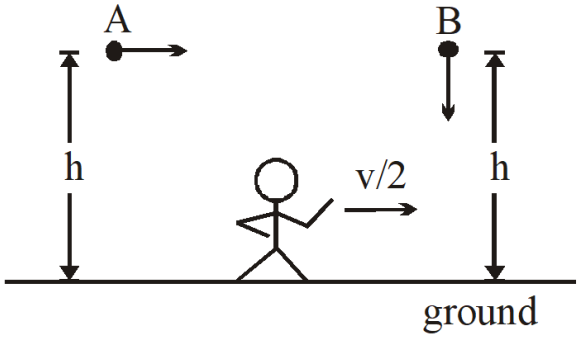
(1) $\frac{g}{2}\left[\frac{n(gn/2-u)}{(gn-u)}\right]^2$

(2) $\frac{g}{2}\left[\frac{n(gn-u/2)}{(gn-u)}\right]^2$

(3) $g\left[\frac{(gn-u)}{(gn-u/2)}\right]^2$

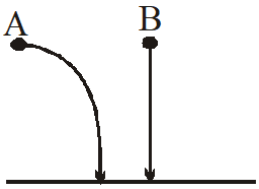
(4) None of these

24) Two identical balls are set into motion simultaneously from equal heights h. While the ball A is thrown horizontally with velocity v, the ball B is just released to fall by itself. Choose the alternative that best represents the motion of A and B with respect to an observer who moves with velocity v/2

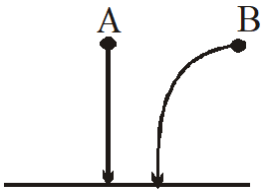


with respect to the ground as shown in the figure.

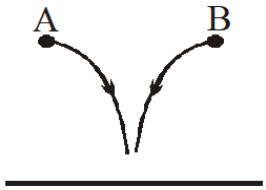
(1)



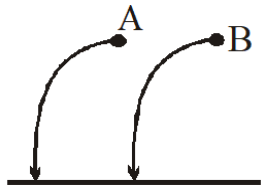
(2)



(3)



(4)



25) What is the length of projection of $\vec{A} = 3\hat{i} + 4\hat{j} + 5\hat{k}$ on xy plane?

(1) 5

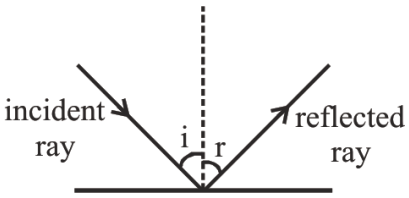
(2) 3

(3) $5\sqrt{2}$

(4) 4

26) Law of reflection states that when a light ray is incident on a reflecting surface then angle of incidence (i) = Angle of reflection(r).

If incident ray is given by $\left[\frac{\hat{i}}{2} + \frac{\sqrt{3}}{2}\hat{j}\right]$ and reflected ray is given by $\left[\frac{\hat{i}}{2} - \frac{\sqrt{3}}{2}\hat{j}\right]$, then angle of incidence is :-



(1) 30°

(2) 45°

(3) 60°

(4) 120°

27) A boat on south bank of river wants of cross a river flowing eastward with speed 3 m/s. The wind is also flowing towards east. If the boat crosses the river in minimum time, the flag on boat points towards south. Find the speed of wind, if speed of boat in still water is 4 m/s :-

(1) 5 m/s

(2) 4 m/s

(3) 2 m/s

(4) 3 m/s

28) A man can swim in still water with speed v wants to cross the river of width d, flowing with speed u(>v) then calculate time to cross the river ..... with minimum drift :-

(1) $\frac{d}{\sqrt{v^2+u^2}}$

(2) $\frac{dv}{u\sqrt{v^2-u^2}}$

(3) $\frac{du}{v\sqrt{u^2-v^2}}$

(4) $\frac{2d}{\sqrt{u^2+v^2}}$

29) At a given instant, an observer stationary on the ground sees a package falling with speed v_1 at certain angle with the vertical. A pilot flying at a constant horizontal velocity relative to the ground sees the package falling vertically with a speed v_2 at the same instant. What is the speed of the pilot relative to the ground ?

(1) $v_1 + v_2$

(2) v1 – v2

(3) $\sqrt{v_1^2-v_2^2}$

(4) $\sqrt{v_1^2+v_2^2}$

30) If $y = \frac{x^2}{(x+1)}$ then $\frac{dy}{dx}$ is :-

(1) $\frac{x(3x+2)}{(x+1)^2}$

(2) $\frac{-x(x+2)}{(x+1)^2}$

(3) $\frac{x(x+2)}{(x+1)^2}$

(4) $\frac{x(x+2)}{(x+1)}$

31) A particle moves with uniform acceleration and v_1 , v_2 and v_3 denote the average velocities in the three successive intervals of time t_1, t_2 and t_3. Which of the following relations is correct ?

(1) $(v_1 - v_2) : (v_2 - v_3) = (t_1 - t_2) : (t_2 + t_3)$

(2) $(v_1 - v_2) : (v_2 - v_3) = (t_1 + t_2) : (t_2 + t_3)$

(3) $(v_1 - v_2) : (v_2 - v_3) = (t_1 - t_2) : (t_1 - t_3)$

(4) $(v_1 - v_2) : (v_2 - v_3) = (t_1 - t_2) : (t_2 - t_3)$

32) An object is moving in x–y plane with the position is a function of time is given by $\vec{r} = x(t)\hat{i} + y(t)\hat{j}$. Point 'O' is as origin having x = 0 and y = 0. The object is definitely mainly towards origin 'O' when :-

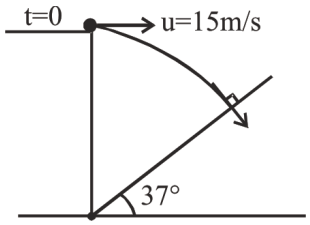
(1) $\frac{dx}{dt} > 0, \frac{dy}{dt} > 0$

(2) $\frac{dx}{dt} < 0, \frac{dy}{dt} < 0$

(3) $x\frac{dx}{dt} < 0, y\frac{dy}{dt} < 0$

(4) $x\frac{dx}{dt} > 0, y\frac{dy}{dt} > 0$

33) A particle is thrown horizontally with speed 15 m/s. Find the time at which it strikes the inclined plane perpendicularly as shown in figure. ($g = 10\ m/s^2$)



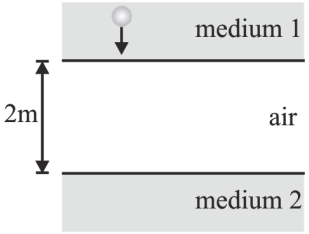
(1) 2 sec

(2) 4 sec

(3) 1 sec

(4) 10 sec

34) A bead is falling in medium 1 with a uniform speed of 3 m s^{-1} for 2s. Then it enters into air and falls freely under gravity for 2 m. Finally, it enters medium 2 and immediately moves with uniform speed for 3 s. Find the total distance the bead has traveled. ($g = 10\ m/s^2$) :-



(1) 27 m

(2) 18 m

(3) 29 m

(4) 23 m

35) The velocity of a particle moving along the x-axis is given by $v = v_0 + \lambda x$, where v_0 and λ are constant. Find the velocity in terms of t :-

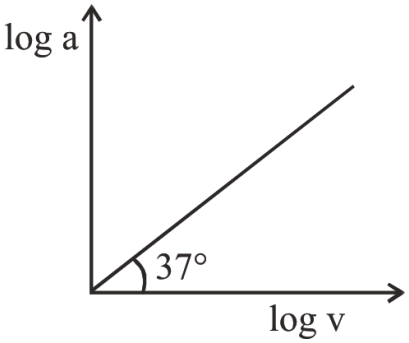
(1) $v = v_0 e^{-\lambda t}$

(2) $v = v_0 e^{\lambda t}$

(3) $v = (v_0 + \lambda) e^{\lambda t}$

(4) $v = (v_0 - \lambda) e^{-\lambda t}$

36) Diagram shows the plot of velocity and aceeleration on log-log scale. If initial velocity is zero, then find relation between velocity and time :-



(1) $v = \left(\frac{t}{4}\right)^4$

(2) $v = \log t$

(3) $v = 2\sqrt{t}$

(4) $v = 4\sqrt{t} v = 4\sqrt{t}$

37)

A particle, initially at rest, starts moving in a straight line with an acceleration a = (6t + 4) m/s^2. The distance covered by it in 3s is :-

(1) 15 m

(2) 30 m

(3) 45 m

(4) 60 m

38) Assertion :- Two persons A and B jump from same building in a time internal of Dt, still one person observe the other person moving with constant speed.
Reason :- They are constantly moving apart from each other.

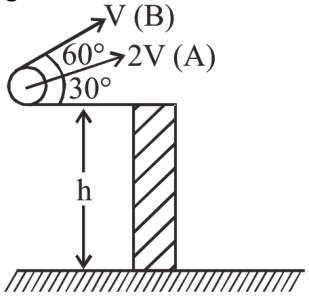
(1) Both Assertion & Reason are correct but Reason is not the correct explanation of Assertion

(2) Both Assertion and Reason is not correct

(3) Assertion is not correct but Reason is correct

(4) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion

39) Two particles A & B are projected from a building. As shown in figure. Which particle will hit the ground earlier :-



(1) Particle A

(2) Particle B

(3) Both at same time

(4) None

40) A girl riding a bicycle with a speed of 5 ms^{-1} towards North direction seas raindrops falling vertically downwards. On increasing the speed to 15 ms^{-1} rain appears to fall making an angle of 45° of the vertical. Find the magnitude of velocity of rain :-

(1) 5 ms^{-1}

(2) $5\sqrt{5}$ ms^{-1}

(3) 25 ms^{-1}

(4) 10 ms^{-1}

41) A projectile is fired from the surface of the earth with a velocity of 5 ms^{-1} at angle θ with the horizontal. Another projectile fired from another planet with a velocity of 3 ms^{-1} at the same angle follows a trajectory of the projectile fired from the earth. The value ofthe acceleration due to gravity on the planet is (in ms^{-2}) is
(given, g = 9.8 ms^{-2}) :-

(1) 3.5

(2) 5.9

(3) 16.3

(4) 110.8

42) The relation between time t and distance x is $t = ax^2 + \beta x$ where α and β are constants. The retardation is

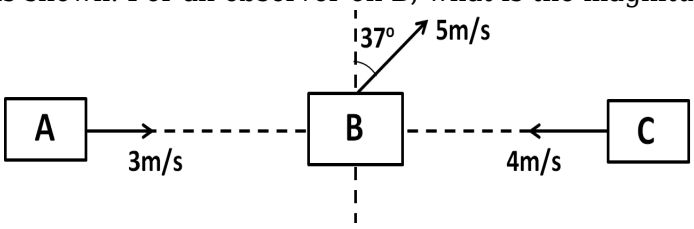
(1) $2\alpha v^3$

(2) $3\beta v^3$

(3) $2\alpha\beta v^3$

(4) $2b^2 v^3$

43) Consider the motion of three bodies as shown. For an observer on B, what is the magnitude of



relative velocity of A with respect to C ?

(1) 10 m/s

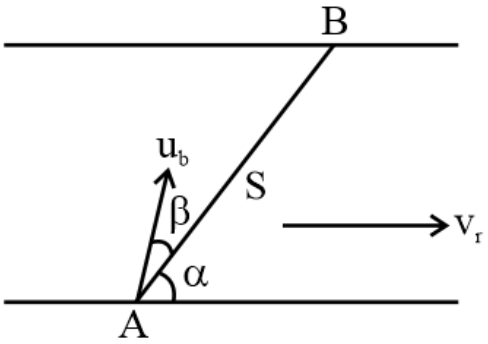
(2) 4 m/s

(3) 12 m/s

(4) 7 m/s

44) A boat travel between two points A and B on the opposite banks of river (as shown), always following the line AB. The distance S between A and B is 240 m. The velocity of river flow $v_r = 2\sqrt{2} m/s$ is constant over the entire width of the river. The line AB makes an angle $\alpha = 45^\circ$ with the direction of the current. The boat moves to cover the distance AB and back in a $t = 2\sqrt{3}$

time min. The angle β remains the same during the passage from A to B and from B to A. At what angle β to the line AB should the boat travel



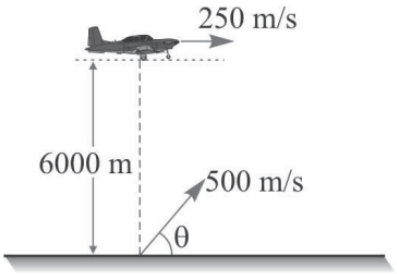
(1) 30^0

(2) 0^0

(3) 15^0

(4) 23^0

45) An aircraft moving with a speed of 250 m/s is at a height of 6000 m, Must overhead of an anti aircraft gun. If the muzzle velocity is 500 m/s, the firing angle θ should be:



(1) 30°

(2) 45°

(3) 60°

(4) none of these.

CHEMISTRY

1)

The equivalent weight of $S_2O_3^{2-}$. When it act as oxidising agent $S_2O_3^{-2} \rightarrow H_2S + SO_4^{-2}$

(1) M/8

(2) M/2

(3) M/4

(4) M/6

2)

$KMnO_4$ reacts with ferrous ammonium sulphate according to the equation
$MnO_4^- + 5Fe^{2+} + 8H^+ \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O$, here 10 ml of 0.1 M $KMnO_4$ is equivalent to

(1) 20 ml of 0.1 M $FeSO_4$

(2) 30 ml of 0.1 M $FeSO_4$

(3) 40 ml of 0.1 M $FeSO_4$

(4) 50 ml of 0.1 M $FeSO_4$

3) A solution of sulphur dioxide in water reacts with H_2S precipitating sulphur. Here sulphur dioxide acts as :-

(1) As oxidising agent

(2) A reducing agent

(3) An acid

(4) A catalyst

4)

In which one of the following reactions, nitrogen is NOT reduced ?

(1) $NO_2 \rightarrow NO_2^-$

(2) $NO_3^- \rightarrow NO$

(3) $NO_3^- \rightarrow NH_4^+$

(4) $NH_4^+ \rightarrow N_2$

5) A compound contain atoms X, Y, Z. The oxidation number of X is +2, Y is +5 and Z is –2. The possible formula of the compound is :-

(1) XY_1Z_2

(2) $Y_2(XZ_3)_2$

(3) $X_3(YZ_4)_2$

(4) $X_3(Y_4Z)_2$

6) A_2O_x is oxidised to AO_3^- by MnO_4^- in acidic medium. If 1.5×10^{-3} mole of A_2O_x requires 40 ml of 0.03M $KMnO_4$ solution in acidic medium. Which of the following statement(s) is/are correct?

(1) The value of "x" = 1

(2) The value of "x" = 3

(3) Empirical formula of oxide is AO_3

(4) Empirical formula of oxide is A_2O

7) In which of the following reactios, the equivalent mass of the underlined species is equal to its molecular mass ? correct code is :-

(a) $\underline{H_3PO_4} + Ca(OH)_2 \rightarrow CaHPO_4 + 2H_2O$

(b) $I_2 + \underline{2Na_2S_2O_3} \rightarrow Na_2S_4O_6 + 2NaI$

(c) $\underline{2Cu^{2+}} + 4I^- \rightarrow Cu_2I_2 + I_2$

(d) $\underline{2NH_2OH} + 4Fe^{3+} \rightarrow N_2O + H_2O + 4Fe^{2+} + 4H^+$

(1) a, c

(2) b, c

(3) a, c, d

(4) a, b, d

8) In an acidic solution, I^- changes to I_2. How many grams of I_2 is produced if in the same process, 1.5×10^{22} electrons are used upto reduce H_3AsO_4 to H_3AsO_3 ? (I = 127)

(1) 1.6 g

(2) 6.4 g

(3) 4.8 g

(4) 3.2 g

9) The oxidation number of C in HCN and HNC respectively are :-

(1) +2, +2

(2) +2, +4

(3) +4, +4

(4) –2, –2

10) The reaction is balanced if:-
$5H_2O_2 + xBrO_2 + 2OH^- \longrightarrow xBr^- + yO_2 + 6H_2O$;

(1) x = 5, y = 2

(2) x= 2, y = 5

(3) x = 4, y = 10

(4) x = 5, y = 5

11) In the following conversion of sulphide of phosphorous
$P_4S_3 \rightarrow P_2O_5 + SO_2$
Equivalent weight of P_4S_3 (molecular weight = M) is :

(1) $\frac{M}{14}$

(2) $\frac{M}{18}$

(3) $\frac{M}{32}$

(4) $\frac{M}{38}$

12) Ratio of number of oxygen atoms which show –1 and –2 oxidation states in the compound $H_2S_2O_8$ is

(1) 1 : 1

(2) 1 : 3

(3) 3 : 1

(4) None of these

13)

Equivalent mass of $Na_2S_2O_3$ during conversion into $Na_2S_4O_6$ is same as :-
(Atomic masses : K = 39, Mn = 55, Na = 23)

(1) Equivalent mass of $KMnO_4$ as oxidant in acidic medium

(2) Equivalent mass of $KMnO_4$ as oxidant in basic medium

(3) Equivalent mass of $KMnO_4$ as oxidant in neutral medium

(4) Equivalent mass of $KMnO_4$ in weakly alkaline medium

14) Equivalent weight of H_3PO_2 when it dispropornate in to PH_3 and H_3PO_3 is-

(1) M

(2) M/2

(3) M/4

(4) 3M/4

15) In $[Cr(O_2)\ (NH_3)_4\ H_2O]Cl_2$; oxidation number of Cr is + 3, then oxygen will be in the form:-

(1) oxide

(2) Peroxide

(3) Superoxide

(4) none

16) Difference between two types of sulphur atoms in $Na_2S_4O_6$ will be :-

(1) 5

(2) 3

(3) 2.5

(4) 6

17) When $KBrO_3$ ion reacts with Br^- ion in acid solution Br_2 is liberated. The equivalent weight of $KBrO_3$ in this reaction is :-

(1) M/8

(2) M/3

(3) M/5

(4) M/6

18) Match the column :-

	Column I (Reduction Process)		**Column II (Charge Required)**
(a)	1 mol of MnO_4^- to Mn^{2+}	(p)	193000C

(b)	1 mole of $Cr_2O_7^{2-}$ to Cr^{3+}	(q)	289500C
(c)	1 mole of Sn^{4+} to Sn^{2+}	(r)	482500C
(d)	1 mole of Al^{3+} to Al	(s)	579000C

(1) a → p, b → r, c → q, d → s

(2) a → r, b → s, c → q, d → p

(3) a → s, b → q, c → p, d → r

(4) a → r, b → s, c → p, d → q

19) The oxidation states of S-atoms in Caro's and Marshall's acids are :-

(1) +6, +6

(2) +6, +4

(3) +6, –6

(4) +4, +6

20) Equivalent wt. of H_3PO_4 in each of the reaction will be respectively :

$$H_3PO_4 + OH^- \rightarrow H_2PO_4^- + H_2O$$
$$H_3PO_4 + 2OH^- \rightarrow HPO_4^{2-} + 2H_2O$$
$$H_3PO_4 + 3OH^- \rightarrow PO_4^{3-} + 3H_2O$$

(1) 98, 49, 32.67

(2) 49, 98, 32.67

(3) 98, 32.67, 49

(4) 32.67, 49, 98

21) In the reaction, $P_2H_4 \rightarrow PH_3 + P_4H_2$, the equivalent mass of P_2H_4 may be given as:

(1) $\frac{3m}{4}$

(2) $\frac{5m}{6}$

(3) $\frac{3m}{5}$

(4) $\frac{12m}{13}$

22) For a reaction $Fe_{0.9}O \rightarrow Fe_2O_3$. What is the equivalent weight of $Fe_{0.9}O$: [molar mass = M]

(1) $\frac{M}{0.8}$

(2) $\frac{M}{0.7}$

(3) $\frac{M}{0.9}$

(4) $\frac{M}{0.85}$

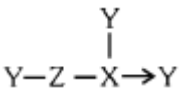
23)

In a conjugate pair of reductant and oxidant, the oxidant has

(1) Higher oxidation number
(2) Lower oxidation number
(3) Same oxidaiton number
(4) None of these

24)

The oxidation number of X in the following structure is :- (Order of E–N is x < y < z)



(1) +2
(2) +4
(3) +1
(4) +3

25) The oxidation states of the most electronegative element in the products of the reaction of BaO_2 with dilute H_2SO_4.

(1) 0 and –1
(2) –1 and –2
(3) –2 and 0
(4) –2 and +2

26)

1 mole of FeC_2O_4 is oxidised by x mole $Cr_2O_7^{-2}$ in acidic medium x is ?

(1) 1
(2) 1.5
(3) 3
(4) 0.5

27) Identify the set from the following sets in which all species can exhibit disproportionation reactions.

(1) $Cl_2, ClO_2^-, ClO_3^-, S_8$
(2) $ClO_4^-, ClO^-, ClO_2^-, F_2$
(3) $ClO_3^-, ClO_4^-, H_2O_2, ClO^-$
(4) $ClO_2^-, ClO_3^-, ClO_4^-, Cl_2$

28)

$C_2H_6(g) + nO_2 \rightarrow CO_2(g) + H_2O\ (l)$

In this equation, the number of electrons involved per mole of ethane is

(1) 28
(2) 14
(3) 7
(4) 21

29)

When a metal is burnt, its weight is increased by 24%. The equvalent weight of the metal will be :-

(1) 25
(2) 24
(3) 33.3
(4) 76

30) One mole of P_2H_4 loses ten moles of electrons to form a new compound Y. Assuming that all the phosphorous appears in the new compound, what is the oxidation state of phosphorous in Y ?

(1) –1
(2) –3
(3) +3
(4) +5

31) A compound of Xe and F is found to have 53.5% Xe. What is the oxidation number of Xe in this compound? (At. wt. of Xe = 131) :-

(1) –4
(2) +2
(3) +4
(4) +6

32) In the reaction, $CrO_5 + SnCl_2 \rightarrow CrO_4^{-2} + SnCl_4$, the element undergoing oxidation and reduction respectively are :-

(1) Cr, Sn
(2) Sn, Cr
(3) Sn, O
(4) Cl, Cr

33) 0.1 M solution of $KMnO_4$ (in acidic medium) may oxidise (having same volume)

(1) $0.25\ M\ C_2O_4^{-2}$
(2) $0.8\ M\ Fe^{+2}$
(3) $0.1\ M\ FeC_2O_4$
(4) $0.6\ M\ Cr_2O_7^{-2}$

34) 25 ml of a 0.1 M solution of a cation of transition metal Z reacts exactly with 25 ml of 0.04 M acidified $KMnO_4$ solution. Which of the following is most likely to represent the change in oxidation state of Z ?

(1) $Z^{+} \rightarrow Z^{2+}$

(2) $Z^{2+} \rightarrow Z^{3+}$

(3) $Z^{3+} \rightarrow Z^{4+}$

(4) $Z^{2+} \rightarrow Z^{4+}$

35) If M_4O_5 compound is written as $M_x(+1)\ M_y(+3)O_5$ then calculate the value of x & y :-

(1) 1, 5

(2) 2, 2

(3) 3, 1

(4) 1, 3

36) How many moles of hydrazine (N_2H_4) can be oxidized into N_2 using 2/3 mole of bromate ions $(BrO_3^- \rightarrow Br^-)$:-

(1) $\frac{1}{3}$

(2) 1

(3) 1.5

(4) $\frac{2}{3}$

37) In the redox reaction $xMnO + yPbO_2 + zHNO_3 \rightarrow HMnO_4 + Pb(NO_3)_2 + H_2O$

(1) x = 2, y = 5, z = 10

(2) x = 2, y = 7, z = 8

(3) x = 2, y = 5, z = 8

(4) x = 2, y = 5, z = 5

38) 1 mole of equimolar mixture of ferric oxlate and ferrous oxalate will require x mole of $KMnO_4$ in acidic medium for complete oxidation x is :-

(1) 4.5 mole

(2) 1.2 mole

(3) 0.9 mole

(4) 0.5 mole

39) Which of the following reaction is non-redox?

(1) $2NaNO_3 \rightarrow 2NaNO_2 + O_2$

(2) $CaO + SiO_2 \rightarrow CaSiO_3$

(3) $Fe + H_2SO_4 \rightarrow FeSO_4 + H_2$

(4) $4Ag + 8CN^- + O_2 + 2H_2O \rightarrow 4[Ag(CN)_2]^- + 4OH^-$

40) 100 ml of 0.1 N hypo decolourised iodine by the addition of χ g of crystalline copper sulphate to excess of KI. The value of 'χ' is (molecular wt. of $CuSO_4.5H_2O$ is 250)

(1) 5.0 g

(2) 1.25 g

(3) 2.5 g

(4) 4 g

41) 4 mole of a mixture of Mohr's salt and $Fe_2(SO_4)_3$ requires 500 mL of 1 M $K_2Cr_2O_7$ for complete oxidation in acidic medium. The mole % of the Mohr's salt in the mixture is :-

(1) 25

(2) 50

(3) 60

(4) 75

42) When 2.5 g of a sample of Mohr's salt reacts completely with 50 mL of $\frac{N}{10}$ $KMnO_4$ solution. The % purity of the sample of Mohr's salt is(Molecular Mass of Mohr's Salt = 392) :-

(1) 78.40

(2) 80.40

(3) 82.40

(4) 84.40

43) 6×10^{-3} mole $K_2 Cr_2O_7$ reacts completely with 9×10^{-3} mole X^{+n} to give XO_3^- and Cr^{+3}. The value of n is :-

(1) 1

(2) 2

(3) 3

(4) None of these

44) Choose the incorrect statement among the following-

(1) $Br_2 + OH^- \rightarrow Br^- + BrO_3^- + H_2O$,
equivalent weight of Br_2 in this reaction is 96 [at. mass of Br = 80]

(2) $SnCl_2 + HCl + O_3 \rightarrow SnCl_4 + H_2O$,
equivalent weight of O_3 in this reaction is 8 [at. mass of O = 16]

(3) $Fe_2(C_2O_4)_3 \rightarrow Fe^{+++} + CO_2$, equivalent weight of $Fe_2(C_2O_4)_3$ in this reaction is 47 [at. mass of Fe = 56, C = 12, O = 16]

(4) $S_2O_3^{-2} \rightarrow SO_4^{-2}$, equivalent weight of $S_2O_3^{-2}$ in this reaction is 14 [at. mass S = 32, O = 16]

45) An element 'A' in a compound ABD has oxidation number –n. It is oxidised by $Cr_2O_7^{2-}$ in acidic medium. In the experiment
1.68×10^{-3} moles of $K_2Cr_2O_7$ were used for
3.36×10^{-3} moles of ABD. The new oxidation number of A after oxidation is :-

(1) 3

(2) 3 – n

(3) n – 3

(4) + n

BIOLOGY

1) Poleward movement of chromatids occurs in the phase of cell division, also characterised by:-

(1) Study of the shape of chromosome.

(2) Attachment of spindle fibres to centromere.

(3) Assemblage of nuclear envelope around the chromosome cluster.

(4) Reform stage of nucleolus.

2) Centrioles undergo duplication during __(i)__ of __(ii)__ and begin to move towards opposite poles of the cell during __(iii)__ stage of __(iv)__ :-

	(i)	**(ii)**	**(iii)**	**(iv)**
(1)	S-Phase	Interphase	Prophase	Mitosis
(2)	S-Phase	Interphase	Anaphase	Mitosis
(3)	Prophase	Mitosis	Metaphase	Mitosis
(4)	Prophase	Mitosis	Anaphase	Mitosis

(1) 1

(2) 2

(3) 3

(4) 4

3) Following are the events occurs during meiosis-I :-
(A) Appearance of chiasmata
(B) Utilization of recombinase
(C) Pairing of homologous chromosomes
(D) Breakdown of nuclear envelope.
Choose the correct sequence of these events from the following :-

(1) A → B → C → D

(2) C → B → A → D

(3) B → C → D → A

(4) D → C → B → A

4)

Most of the histone production during cell cycle occurs in :

(1) G_0 phase

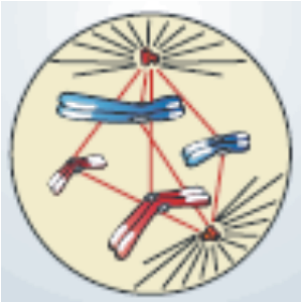
(2) G_1 phase

(3) S phase

(4) G_2 Phase

5)

Choose the correct option for the diagram given below :-



	Number of Chromosomes	Number of Chromatids	Number of Centromeres
(1)	4	8	8
(2)	4	4	4
(3)	4	16	8
(4)	4	8	4

(1) 1

(2) 2

(3) 3

(4) 4

6) How many of the following are correct statements regarding meiosis ?
(a) Meiosis-I is initiated after the parental chromosomes have replicated to produce identical sister chromatids at the s-phase
(b) Meiosis involves pairing of homologous chromosomes and recombination between them
(c) Four haploid cells are formed at the end of meiosis-II
(d) Meiosis ensures the production of haploid phase in life cycle

(1) One

(2) Two

(3) Three

(4) All four

7) Most variable phase of cell cycle is

(1) G_1

(2) G_2

(3) S

(4) M

8) Ends of chromosomes are attached with nuclear envelope at attachment phase in-

(1) Leptotene

(2) Zygotene

(3) Pachytene

(4) Diplotene

9) If there are 30 chromosome in G_1 phase then what will be number of bivalents in zygotene?

(1) 30

(2) 15

(3) 45

(4) 60

10) Read the following statements and select the correct option that bears true statements :-
(i) During S-phase, amount of DNA per cell doubles, but chromosome number remains same.
(ii) A cell in G_0 phase is metabolically inactive but no longer proliferate unless called on to do so depending on the requirement of the organisms
(iii) Cell division is a progressive process that shows the distinct boundries between various stages.
(iv) At the end of prophase, golgi complex, endoplasmic reticulum, nucleolus and nuclear envelope get disappear

(1) only i, ii and iv

(2) only i, ii and iii

(3) only ii, iii and iv

(4) Only i and iv

11) If in a cell the amount of DNA is 2C in metaphase then what will be the amount of DNA in anaphase (x), in G_1 phase (y) and after M-phase (z) ?

(1) (x) – 4c, (y) – c, (z) – c

(2) (x) – 2c, (y) – 2c, (z) – 4c

(3) (x) – 4c, (y) – 4c, (z) – 2c

(4) (x) – 2c, (y) – c, (z) – c

12) In which one of the following options two events of cell cycle are not correctly matched with their particular phase ?

Events	Phase

(1)	(i)	Centriole duplication in the cytoplasm of animal cell	S
	(ii)	DNA replication in nucleus	
(2)	(i)	Splitting of centromere	Anaphase
	(ii)	Movement of centriole toward opposite poles of cell	
(3)	(i)	Activity of recombinase enzyme	Pachytene
	(ii)	Crossing over between non sister chromatid	
(4)	(i)	Terminalisation of chiasmata	Diakinesis
	(ii)	Nucleolus disappear	

(1) 1

(2) 2

(3) 3

(4) 4

13) Longitudinal duality of each chromosome of a homolo-gous pair becomes visible at

(1) Pachytene

(2) Diakinesis

(3) Leptotene

(4) Anaphase

14) If a spore mother cell has 10 pg DNA in G_1 stage then how much present in a spore after meiosis?

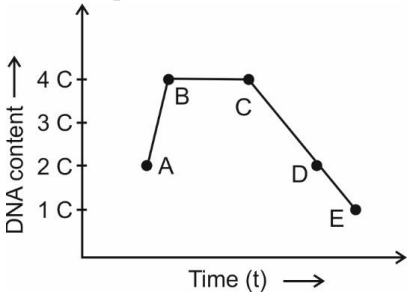
(1) 10 pg

(2) 5 pg

(3) 2.5 pg

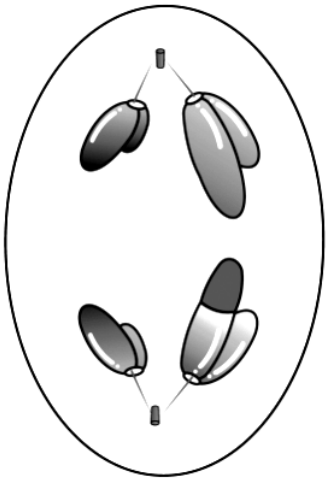
(4) 15 pg

15) Examine the phase from graph given below which shows the change in amount of DNA during various phases (A-D) of meiotic division.



	A	B	C	D	E
(1)	Meiosis-I	Meiosis-II	G_2	S	G_1
(2)	G_1	S	G_2	Meiosis-I	Meiosis-II
(3)	S	G_2	Meiosis-I	G_1	Meiosis-II
(4)	G_2	Meiosis-I	S	G_1	Meiosis-I

(1) 1
(2) 2
(3) 3
(4) 4



16) Identify given stage in figure :-

(1) Telophase - II
(2) Anaphase - II
(3) Metaphase
(4) Telophase

17) Yeast cell divides once in approximately every -

(1) 90 minutes
(2) 9 minutes
(3) 24 hours
(4) 24 days

18) Number of mitotic division required for formation of 128 cells :

(1) 127
(2) 32
(3) 12
(4) 64

19) Match the phases of Prophase I in Meiosis I with their key characteristics:

	Column A (Phase)		**Column B (Characteristic)**
1	Leptotene	A	Chromosomes become visible and start condensing
2	Zygotene	B	Homologous chromosomes pair and synaptonemal complex forms
3	Pachytene	C	Crossing over occurs, recombination nodules appear
4	Diplotene	D	Dissolution of synaptonemal complex and chiasmata form
5	Diakinesis	E	Terminalisation of chiasmata, spindle assembly begins

(1) 1-A, 2-B, 3-C, 4-D, 5-E

(2) 1-B, 2-C, 3-E, 4-A, 5-D

(3) 1-C, 2-D, 3-A, 4-E, 5-B

(4) 1-E, 2-A, 3-B, 4-C, 5-D

20) A characteristic feature of meiosis is

(1) Two DNA replications and one division of cytoplasm

(2) A single DNA replication and four divisions of cytoplasm

(3) Two DNA replication cycles and two divisions of cytoplasm

(4) A single DNA replication and two divisions of cytoplasm

21) Synapsis occurs between :

(1) Spindle fibres and centromere

(2) Two homologous chromosomes

(3) A male and a female gamete

(4) mRNA and ribosomes

22) Nuclear envelope disappears at

(1) metaphase

(2) anaphase

(3) early prophase

(4) late prophase

23) Bivalents are formed in

(1) diplotene
(2) pachytene
(3) zygotene
(4) interkinesis

24) If a cell have 16 tetrads in G_1 phase then find out the numbers of chromatids in metaphase-I and metaphase-II respectively ?

(1) 16 & 32
(2) 32 & 64
(3) 64 & 32
(4) 32 & 32

25) Select the incorrect statements :
(A) During prophase, initiation of the assembly of mitotic spindle.
(B) Synptonemal complex formation during diplotene.
(C) X-shaped structures (chiasmata) are formed during diakinesis.
(D) Homologous chromosome separation during anaphase-II.
(E) Crossing over is an enzyme mediated process and enzyme involved is called recombinase.
Choose the correct answer from the options given below :-

(1) B and C only
(2) A, C and D only
(3) C, D and E only
(4) B, C and D only

26) **Assertion:-** Sister chromatids remain associated at their centromere in anaphase-I.
Reason:- G_0 phase is metabolically active

(1) Both Assertion & Reason are correct but Reason is not the correct explanation of Assertion
(2) Assertion is correct but Reason is not correct
(3) Assertion is not correct but Reason is correct
(4) Both Assertion and Reason are correct and Reason is the correct explanation of Assertion

27) How many of the following statements are correct with respect to prophase:
A) Chromosomal material condensation
B) Movement of centrioles to opposite poles
C) Nucleolus disappear
D) Spindle fibres start appearing
E) Nuclear envelope disappears

(1) 2
(2) 4
(3) 5
(4) 3

28) Tetrad is made of-

(1) four nonhomologous chromatids
(2) four nonhomologous chromosomes
(3) four homologous chromosomes with four chromatids
(4) two homologous chromosomes and each with two chromatids

29) A pentaploid nucleus cannot undergo meiosis because :-

(1) The DNA cannot replicate
(2) Not all of the chromosomes can form homologous pairs
(3) The sister chromatids cannot separate
(4) Cytokinesis cannot occur

30) Diplotene stage of meiosis

(1) Involves crossing over between homologous chromosome
(2) Can last for months or years in the oocytes of some vertebrates
(3) Involves synapsis of homologus chromosomes
(4) Represents transition to metaphase I

31)

I. Cells like heart cells and nerve cells, which are also called non-dividing cells are in. ...A... phase.

II. During S-phase in animal cells. DNA replication takes place in the ...B... and duplication of centriole takes place in the. ...C... Fill in the blanks with appropriate options.

(1) A-G_0, B - nucleus, C- cytoplasm
(2) A - S, B - cytoplasm, C - nucleus
(3) A-G_1, B - nucleus, C- cytoplasm
(4) A -G_0, B - cytoplasm, C - nucleus

32) Many cells in adult animals exit .....'A'..... to enter an inactive stage called .....'B'..... Cells in this stage remain metabolically .....'C'..... and.....'D'.....
Choose the correct match :-

(1) A = Karyokinesis, B = G_0 phase, C = inactive, D = divide
(2) A = G_1 phase, B = G_0 phase, C = inactive, D = not divide
(3) A = Cytokinesis, B = Polyteny, C = active, D = divide
(4) A = G_1 phase, B = Quiescent stage, C = active, D = not divide

33) Choose the incorrect statements :

(1) M-phase is the most dramatic period of cell cycle
(2) The metacentric chromosome has middle centromere forming two equal arms of the chromosome.
(3) The content of nucleolus is continuous with the rest of the nucleoplasm

(4) The ribosomes of the chloroplasts are smaller than the mitochondrial ribosomes.

34)

In the pachytene, each bivalent contains _____ chromatids.

(1) Two
(2) Four
(3) Eight
(4) Many

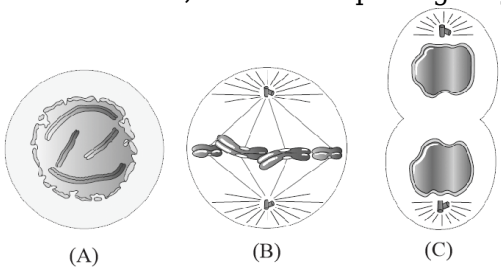
35) Amitosis is common in :-

(1) Root meristem
(2) Eukaryotic cells
(3) Prokaryotic cells
(4) Endosperm

36) During which stage of mitosis, the chromosomes move towards the pole?

(1) Telophase
(2) Anaphase
(3) Prophase
(4) Interphase

37) The figure below shows 3 phases of mitosis, select the option giving correct identification



together with the correct event ?

(1) C-Telophase-Nuclear envelope assembles around the chromosome clusters.
(2) B-Anaphase-Segregation of homologous chromosomes
(3) A-Prophase-Chromosomes get fully condensed
(4) C-Metaphase-Condensation of chromatin to form chromosome

38) Syncytium formation is due to:

(1) Failure of cytokinesis after karyokinesis
(2) Failure of both cytokinesis and karyokinesis
(3) Occurrence of cytokinesis and karyokinesis
(4) Occurrence of karyokinesis after cytokinesis

39) Progression of the cell cycle is regulated by concentration of which type of molecule?

(1) Centrosomes

(2) Cyclin-independent kinases

(3) Cyclins

(4) Microtubules

40) **Assertion :** In higher plant cells, centrioles are present.
Reason : In plant cell division, spindle fibre formed by centrioles.
Choose correct option :-

(1) Both assertion and reason are true

(2) Assertion is true but reason is false

(3) Both assertion and reason are false

(4) Assertion is false but reason is true

41) **Assertion :** We come across meiosis during gametogenesis in animals, this leads the formation of haploid gametes.
Reason : Crossing over leads to recombination of genetic material on the two homologous chromosomes.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

42) Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**
Assertion (A) : The beginning of diplotene is recognised by the dissolution of the synaptonemal complex and formation of X shaped structures called chiasmata.
Reason (R) : In oocytes of some vertebrates, diplotene can last for months or years.
In the light of the above statements, choose the correct answer from the options given below :

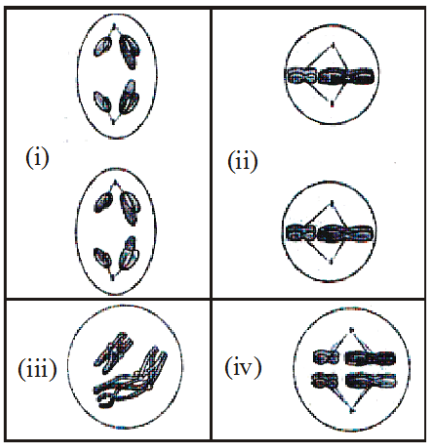
(1) **(A)** is not correct but **(R)** is correct

(2) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

(3) Both **(A)** and **(R)** are correct but **(R)** not the correct explanation of **(A)**

(4) **(A)** is correct but **(R)** is not correct

43) Arrange the different stages of meiosis (i), (ii), (iii) and (iv) in sequence :-



(1) (i) → (iv) → (ii) → (iii)

(2) (iii) → (iv) → (ii) → (i)

(3) (iv) → (iii) → (ii) → (i)

(4) (ii) → (i) → (iii) → (iv)

44) Anaphase is characterized by

(1) Splitting of centromere

(2) Separation of chromatids

(3) Movement of chromatid to opposite pole

(4) All of these

45) Anaphase Promoting Complex (APC) is a protein degradation machinery necessary for proper mitosis of animal cells. If APC is defective in a human cell, which of the following is expected to occur?

(1) Chromosomes will not condense

(2) Chromosomes will be fragmented

(3) Chromosomes will not segregate

(4) Recombination of chromosome arms will occur.

46) Correct sequence of abundance of compounds in the cell is :-

(1) Water > Lipid > carbohydrate > Protein > Ions

(2) Lipid > Nucleic acid > Protein > Carbohydrate > Ions

(3) Water > Protein > Nucleic acid > Carbohydrate > Ions

(4) Ions > Protein > Nucleic acid > Carbohydrate > lipid

47) Polyunsaturated fatty acids are

(1) Oleic acid, linoleic acid and linolenic acid

(2) Oleic acid, palmitic acid and arachidonic acid

(3) Linoleic acid, linolenic acid and stearic acid

(4) Linoleic acid, linolenic acid and arachidonic acid

48) Match the list-I with list-II.

	List-I		**List-II**
(a)	N-base	(i)	Zwitter ion form
(b)	Amino acid	(ii)	Lipid
(c)	Lecithin	(iii)	Fatty acid
(d)	Arachidonic acid	(iv)	Genetic material

(1) a-iv, b-i, c-ii, d-iii

(2) a-i, b-iv, c-iii, d-ii

(3) a-ii, b-i, c-iv, d-iii

(4) a-iv, b-iii, c-i, d-ii

49) Statement-1: Living state is a constant effort to prevent falling into equilibrium.
Statement-2: Living state is a non-equilibrium non-steady state.

(1) Statement-1 is correct, statement-2 is incorrect

(2) Statement-1 and statement-2 are incorrect

(3) Statement-1 and statement-2 are correct

(4) Statement-1 and statement-2 are incorrect

50) There are some nucleic acids that behaves like enzymes and are called-

(1) DNase

(2) RNase

(3) Endonuclease

(4) Ribozymes

51) Consider the following four statements (a-d) and select the option which includes all the correct ones only :-
(a) Lipids are generally water insoluble.
(b) The R group could be a methyl (–CH_3) or ethyl (–C_2H_5) or higher no. of –CH_2 group (1 carbon to 19 carbon) in fatty acids
(c) Palmitic acid has 16 carbons including the carboxyl carbon.
(d) Arachidonic acid has 21 carbon atom including the carboxyl carbon.
Option

(1) Statement (a), (b) and (c)

(2) Statement (a), (b) only

(3) Statement (c), (d) only

(4) Statement (a), (c) and (d)

52) Sweetest sugar in nature is :-

(1) Fructose

(2) Sucrose
(3) Maltose
(4) Galactose

53) N-Glycosidic bond is found in :-

(1) Glycogen
(2) Cellulose
(3) Starch
(4) DNA

54) Which of the following is an essential amino acid ?

(1) Cysteine
(2) Tyrosine
(3) Tryptophan
(4) Glycine

55) Select the correct carbohydrate according to their catagory :-

	(A) Monosaccharide	(B) Disaccharide	(C) Polysaccharide
(1)	Ribose	Lactose	Insulin
(2)	Glucose	Fructose	Pectin
(3)	Arabinose	Glucose	Chitin
(4)	Fructose	Sucrose	Starch

(1) (1)
(2) (2)
(3) (3)
(4) (4)

56) Read the given statements :-
I. Amino acids are linked by peptide bond
II. Monosaccharides are linked together by glycosidic bond
III. Lecithin is a component of cell membrane
IV. RuBisCO is the most abundant protein in animal world
Which of the given statement are correct

(1) I, II, III & IV
(2) I, & II
(3) I, II & III
(4) I, II & IV

57) Match the column-I with column-II and choose correct option :-

	Column-I		Column-II
A	Insulin	i	Fights infectious agents
B	Antibody	ii	Enables glucose transport into cells
C	Receptor	iii	Hormone
D	GLUT-4	iv	Sensory reception

(1) A-iii, B-ii, C-iv, D-i

(2) A-iii, B-i, C-iv, D-ii

(3) A-i, B-ii, C-iii, D-iv

(4) A-ii, B-iii, C-iv, D-i

58) Match the column-

Column I		Column II	
A.	Toxins	i.	Morphine
B.	Alkaloids	ii.	Abrin
C.	Lectins	iii.	Vinblastin
D.	Drugs	iv.	Concanavalin A

	A	B	C	D
1	i	ii	iii	iv
2	i	iii	ii	iv
3	ii	i	iv	iii
4	iv	ii	iii	i

(1) 1

(2) 2

(3) 3

(4) 4

59) N-acetyl galactosamine is ?

(1) Modified polysaccharide

(2) Modified sugar

(3) Derivative of protein

(4) Amino acid

60) In triglyceride one molecule of Glycerol reacts with ___A___ molecule of fatty acids, While ___B___ molecules of fatty acids are present in Lecithin.
In above paragraph A and B are respectively?

(1) 2,3

(2) 3,2

(3) 2,2,

(4) 1,1

61) Which of the following is aromatic amino acid?

(1) Valine

(2) Tryptophan

(3) Tyrosine

(4) Both (2) and (3)

62) The acid insoluble fraction of the tissue has

(1) Amino acids, Proteins and nucleic acids

(2) Monosaccharides and polysaccharides

(3) Nucleic acids, nucleotides and nitrogen base

(4) Proteins, nucleic acids, polysaccharides and lipids

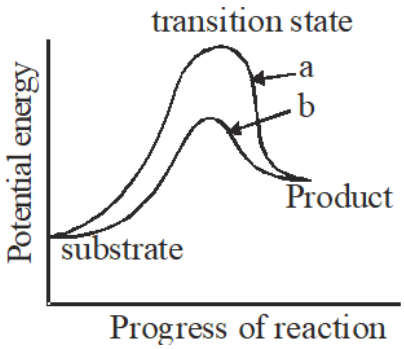
63) Which of the following statements regarding cofactors is not correct ?

(1) Every coenzyme is a cofactor and every cofactor is a coenzyme

(2) Catalytic activity is lost when the cofactor is removed from the enzyme

(3) Cofactor can be organic or inorganic

(4) In the catalase enzyme, haem group work as a cofactor



64)
Choose the incorrect statement(s) regarding above graph :-
(A) This graph is for an exothermic process
(B) The energy difference between product and transition state is less than activation energy.
(C) Graph 'a' shows the presence of enzyme
Options :-

(1) Only B

(2) A and B

(3) B and C

(4) A and C

65) Read the following carefully :
(A) Enzymes are denatured at high temperatures but in certain exceptional organisms they are effective even at temperatures 80–90°C.
(B) Enzyme are highly specific.
(C) Most enzymes are proteins but some are lipids.
(D) Enzymes require optimum pH for maximal activity.
How many is/are wrong :-

(1) Four

(2) Three

(3) Two

(4) One

66)

Most of the digestive enzymes belong to which class of enzyme?

(1) Lyases

(2) Oxidoredutase

(3) Hydrolases

(4) Ligase

67)

Which of the following element is negligible in human body but present in large amounts in Earth's crust :-

(1) Nitrogen

(2) Sulphur

(3) Silicon

(4) Calcium

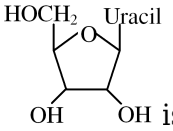
68) Sucrose is composed of :-

(1) α-D glucose and α-D glucose

(2) α-D glucose and α-D fructose

(3) α-D glucose and β-D fructose

(4) α-D fructose and α-D fructose

69) Read the following statements carefully and find out pair of statements which are not correct :-
(a) Some secondary metabolites have ecological importance
(b) NaCl, $CaCO_3$, PO_4^-, CH_3COOH, CH_4 etc are inorganic constituents of living cell
(c) 

is a nucleoside as well as found in RNA along with phosphate
(d) Secondary structure of proteins is absolutely necessary for the many biological activities of proteins

(1) a, b

(2) b, c

(3) c, d

(4) d, b

70) To analyse chemical composition of organic compounds found in living organisms, which of the following is used-

(1) Trichloroacetic acid

(2) Dichloroacetic acid

(3) Dilute hydrochloric acid

(4) Dilute Sulphuric acid

71) Find the incorrect match

	Organic molecule	Example	Monomer	Function
(1)	Lipids	Fats,oil	Fatty acids, Glycerol	Serve as reserve food
(2)	Protein	Actin, GLUT-4 Collagen	Amino acids	Support, transport & regulation
(3)	Nucleic acid	DNA, RNA	Nucleotides	Storage of genetic information
(4)	Carbohydrates	Monosac-Charides Disaccharides, Polysaccharides	Glucose only	Regulatory function

(1) 1

(2) 2

(3) 3

(4) 4

72) In a polysaccharide, right end of chain is A and left end is B.

(1) A - Reducing, B - Reducing

(2) A - Reducing, B - Non-reducing

(3) A - Non-reducing, B-Reducing

(4) A - Non-reducing, B- Non-reducing

73) In an organism DNA, which is double stranded, 17% of the bases were shown to be Cytosine. The percentage of the other three bases expected to be present in this DNA are :-

(1) G-17%, A-16.5%, T-32.5%

(2) G-17%, A-33%, T-33%

(3) G-8.5%, A-50%, T-24.5%

(4) G-34%, A-24.5%, T-24.5%

74) Identify the basic amino acid from the following.

(1) Glutamic Acid

(2) Lysine

(3) Valine
(4) Tyrosine

75) In DNA, the complimentary bases are:

(1) Adenine and guanine, thymine and cytosine
(2) Uracil and adenine; cytosine and guanine
(3) Adenine and thymine; guanine and cytosine
(4) Adenine and thymine; guanine and uracil

76) In the DNA molecule :-

(1) the proportion of Adenine in relation to thymine varies with the organism
(2) there are two strands which run antiparallel one in 5'→3' direction and other in 3'→5'
(3) the total amount of purine nucleotides and pyrimidine nucleotides is not always equal
(4) there are two strands which run parallel in the 5'→3' direction

77) Ligase catalyse :-

(1) Joining of C–O
(2) Oxidation–reduction of substrate
(3) Hydrolysis of C–C
(4) Conversion of optical isomer

78) Arrange the steps of catalytic action of an enzyme in order and choose the right option
(A) The enzyme releases the products of the reaction and the free enzyme is ready to bind to another molecule of the substrate
(B) The active site of the enzyme, now in close proximity of the substrate breaks the chemical bonds of the substrate and the new enzyme-product complex is formed
(C) The binding of the substrate induces the enzyme to alter its shape, fitting more tightly around the substrate
(D) The substrate binds to the active site of the enzyme, fitting into the active site

(1) A → B → C → D
(2) D → B → C → A
(3) D → C → B → A
(4) D → C → A → B

79) How many compounds given in the box below belong to acid insoluble fraction?
Insulin, Chitin, Glucose, Adenine, Lecithin

(1) Five
(2) Four
(3) Three
(4) Two

80) Which of the following is not a component of Lecithin?

(1) Choline
(2) Phosphate group
(3) Sphingosine
(4) Fatty acid

81) How many double bonds are present in oleic acid

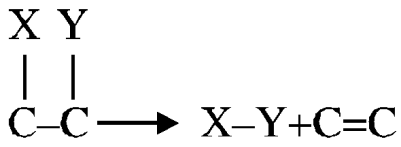
(1) 4
(2) 3
(3) 2
(4) 1

82) In structure the keratin of mammalian hairs is

(1) α-helical like silk protien
(2) β-pleated like silk
(3) α-helical unlike silk fibroin
(4) β-pleated unlike silk

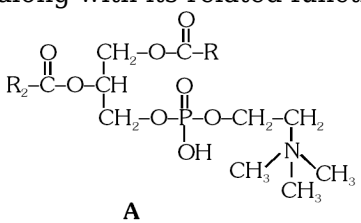
83) Which of the following is not a polymer?

(1) Starch
(2) Nucleic acid
(3) Maltose
(4) Protein

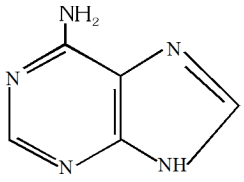
84) Which class of enzyme catalyzes following reaction 

(1) Hydrolase
(2) Ligase
(3) Isomerase
(4) Lyases

85) Which one of the following structural formulae of two organic compounds is correctly identified along with its related function ?



A



B

(1) B : Adenine - A nucleotide that makes up nucleic acids

(2) A : Triglyceride - Major source of energy

(3) B : Uracil - A component of DNA

(4) A : Lecithin - A component of cell membrane

86) Bond between phosphate and sugar in a nucleotide is

(1) H-bond

(2) Covalent bond

(3) Ionic bond

(4) Sulphide bond

87) α-keratin protein assumes secondary structure through the formation of

(1) Intrachain hydrogen bond

(2) Interchain hydrogen bond

(3) Peptide bond

(4) Interchain disulphide bond

88) Non protein organic compound, which shows transient association with apoenzyme usually during the course of catalysis, is known as :-

(1) Coenzyme

(2) Prosthetic group

(3) Metal ion

(4) Pathogen

89) Which of the following is not a fatty acid?

(1) Uridylic acid

(2) Palmitic acid

(3) Stearic acid

(4) Arachidonic acid

90) How many of the statements are **correct** from following :-
(A) Exoskeleton of Arthropods made up of chitin.
(B) N - acetyl - D - Galactosamine is modified sugar.
(C) Starch forms helical structure.
(D) Cellulose is micromolecule.

(1) 1

(2) 2

(3) 3

(4) 4

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	3	2	4	3	3	1	1	2	3	4	2	4	2	1	1	3	4	3	1
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	3	1	3	1	1	4	3	3	3	2	3	1	3	2	1	3	2	2	2
Q.	41	42	43	44	45															
A.	1	1	4	1	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	1	4	1	4	3	2	2	4	1	2	3	2	2	4	3	1	3	4	1	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	2	2	1	2	2	4	1	2	3	3	4	3	1	4	4	2	1	3	2	3
Q.	86	87	88	89	90															
A.	4	1	1	3	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	1	1	2	3	4	4	1	1	2	4	4	2	1	2	2	2	1	1	1	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	2	4	3	3	4	1	3	4	2	2	1	4	4	2	3	2	1	1	3	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	3	2	4	3	3	4	1	1	4	1	1	4	3	(4)	3	2	3	2	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	4	4	1	4	4	3	3	3	4	1	4	2	2	2	3	2	1	3	3	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	4	3	3	4	4	2	1	1	1	3										

SOLUTIONS

PHYSICS

1)

a_x = const.
$a_y = 0$, v_y = const.
At $t = t_0 \Rightarrow v_x = 0$
$\Rightarrow \vec{v} \perp \vec{a}$

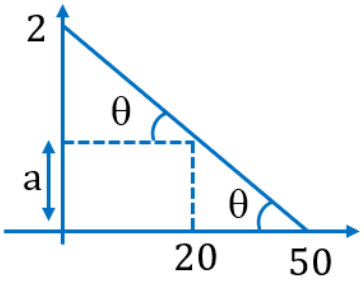
2)

Let t_1 be time taken by top ball to strike ground & t_2 be time taken by bottom ball to strike ground.

$$\Rightarrow (t_1 - t_2) = \frac{2v_0 \sin\theta}{g}$$

Thus, $R_1 - R_2 = (v_0 \cos\theta)(t_1 - t_2)$

$$= \frac{2v_0^2 \sin\theta \cos\theta}{g}$$



3)

$$\tan\theta = \frac{a}{30} \Rightarrow \frac{2}{50}$$

$a \Rightarrow 1.2 m/sec^2$

Area under a-t curve gives is.

$$v_f \Rightarrow \frac{1}{2} \times (1.2 + 2) \times 20$$

$v_f \Rightarrow 32 m/sec$

$v_f \Rightarrow 32 - 10 \Rightarrow 22 m/sec$

$$|\vec{A}| = \sqrt{3^2 + 4^2} = 5$$

$$|\vec{B}| = \sqrt{6^2 + 8^2} = 10$$

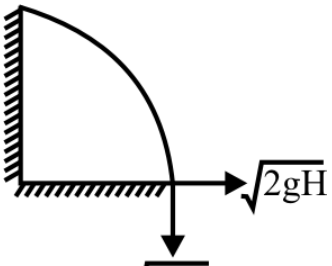
$$\vec{A}.\vec{B} = \left(3\hat{i} + 4\hat{j}\right) . \left(6\hat{i} + 8\hat{j}\right)$$

$$= 18 + 32 = 50$$

4) $$\frac{|\vec{A}|}{|\vec{B}|} = \frac{5}{10} = \frac{1}{2}$$

5)

Component of velocity perpendicular to line AB must be equal.



6) $$v = \sqrt{\left(\sqrt{2gH}\right)^2 + \left(\sqrt{2gH}\right)^2} = 2\sqrt{gH}$$

7)

$$y = x \tan \theta - \frac{gx^2}{2u^2\cos^2\theta}$$

From given initial velocity

$$u = \sqrt{u_x^2 + u_y^2}$$

$$u = \sqrt{1^2 + 2^2} = \sqrt{5}$$

$$\tan \theta = \frac{u_y}{u_x}$$

$$\tan \theta = 2$$

so $$y = 2x - \frac{5x^2}{1} = 2x - 5x^2$$

8) $$a = v\frac{dv}{dx}$$

$$\int adx = \int vdv$$

$$\frac{v^2 - u^2}{2} = \int adx$$ = Area under graph

$\frac{v^2 - 0}{2} = \frac{1}{2} \times 1 \times 1$

v = 1 m/s

9) $\frac{d^2\vec{r}}{dt^2} = \vec{a}$

and $\vec{r}.\vec{a} = 0$

$\Rightarrow t = 1$ sec

10)

$v_{av} = \frac{v_1 + v_2}{2}$ for uniform acceleration/retardation is also velocity at mid time.

In first half time average velocity is higher, so the car travels more than half distance.

11)

For maximum height, $v_y = 0$

At $t = \frac{17}{3}$s , $v_y = 24\left(\frac{17}{3}\right) - 3\left(\frac{17}{3}\right) = \frac{119}{3}$ *m/s*

After that, $a = -g = -10$ *m/s*^{2}

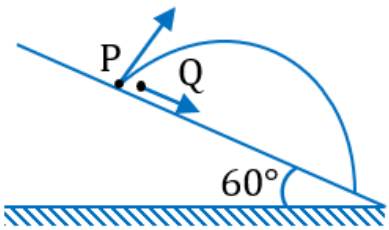
And at $t = t_2$, $v_y = 0$

So $0 = \frac{119}{3} - 10\left(t_2 - \frac{17}{3}\right)$

Or $t_2 = \frac{289}{30} = 9.6$ sec.

12) $t = 4s$

$T = \frac{2u\sin\theta}{g\cos\alpha}$



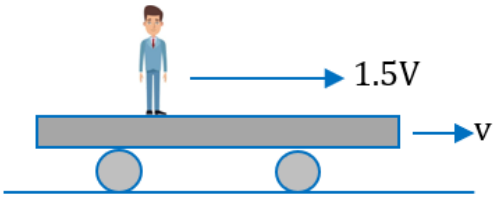
$u = \frac{2u}{10 \times \frac{1}{2}}$

$\frac{20}{2} = u$

$u = 10$ *m/s*

13) $\vec{v}_{m/T} = 1.5v\hat{i}$

$\vec{v}_{T/g} = v\hat{i}$



$x = L$

$u_{m/T} = -1.5v\hat{i}$

$L = ut + \frac{1}{2}at^2$

$t = \frac{2L}{1.5v}$

$D = vt$

$D = \frac{4L}{3}$

14) Relative acceleration of coin w.r.t. lift = 1 m/s^2 ↓

$t = \sqrt{\frac{2h}{a_{rel}}} = \sqrt{\frac{2 \times 2}{1}} = 2$ s

15)

Equation of trajectory is given as

$y = 2x - 9x^2$ ..... (1)

Comparing with equation :

$y = x \tan\theta - \frac{g}{2u^2\cos^2\theta}.x^2$ ..... (2)

We get;

$\tan\theta = 2$

$\Rightarrow \boxed{\cos\theta = \frac{1}{\sqrt{5}}}$

Also, $\frac{g}{2u^2\cos^2\theta} = 9$

$\Rightarrow \frac{10}{2 \times 9 \times \left(\frac{1}{\sqrt{5}}\right)^2} = u^2$

$\Rightarrow u^2 = \frac{25}{9}$

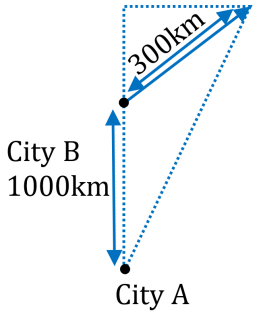
$\Rightarrow \boxed{u = \frac{5}{3}\text{m/s}}$

16) $\frac{1000 + 150\sqrt{2}}{500 + V_g} = 2$

$V_g = 75\sqrt{2}$

$\frac{150\sqrt{2}}{V_n} = 2 \Rightarrow V_x = 75\sqrt{2}$

$V_w = 150$Km/hr $45°N$ of E



17)

Ball strikes towers ⇒ $S_x = 0$

$\therefore 0 = 12 \times t + \frac{1}{2}(-6)t^2 \Rightarrow t = 4$ sec

So displacement in y-direction in 4 sec

$S_y = \frac{1}{2} \times 10 \times 16 = 80m$

18) Slope of v-t curve is constant so acceleration will be constant.
Initially speed decreases and it becomes zero and after that it increases.
Initially velocity in negative direction and then in positive, so it will be change in direction of motion once.

19)

$80 = \frac{1}{2} \times 10 \times t^2 \Rightarrow t = 4$ sec

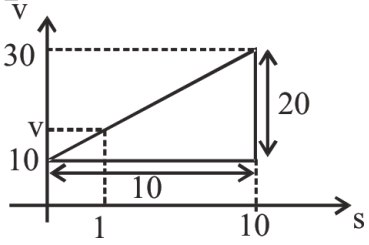
$10 \times 4 + v \times 4 = 60 \Rightarrow v = 5$ m/s

20) $\frac{dv}{ds} = \frac{20}{10} = 2$

$2 = \frac{v - 10}{1}$

$v = 12$

$\therefore a = \frac{v\,dv}{ds} = 12 \times 2 = 24\,m/s^2$



21) $t = ax^2 + bx$

$\frac{dt}{dx} = 2ax + b$

$\Rightarrow \frac{dx}{dt} = \frac{1}{2ax + b} = v$

$$a = \frac{d^2x}{dt} = \frac{d}{dt}(2ax + b)^{-1}$$
$$= -(2ax + b)^{-2} \times \frac{dx}{dt} \times 2a$$
$$= -2a\left(\frac{1}{2ax + b}\right)^2 \times \frac{1}{(2ax + b)}$$
$$= -2a\left(\frac{1}{2ax + b}\right)^3$$
$$= -2av^3$$

22)

Option (3) is correct

23) Let second stone overtakes the first, at time 't' after the first stone was thrown

$\frac{1}{2} gt^2 = u(t - n) + \frac{1}{2}g(t - n)^2$

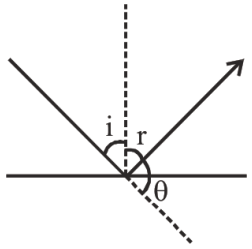
On solving, $t = \left(\frac{un - gn^2/2}{gn - u}\right)$

and required distance $= \frac{1}{2}gt^2$

24)

Option (3) is correct

25) $\ell = \sqrt{3^2 + 4^2} = 5$



26)

$\angle i = \angle r$

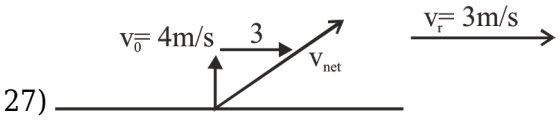
$\cos\theta = \frac{\vec{A}.\vec{B}}{AB} = \frac{\frac{1}{4} - \frac{3}{4}}{(1)\,(1)} = -\frac{1}{2}$

$\theta = 120°$

$i + r + \theta = 180º$

$2i + 120° = 180°$

$i = 30°$



27)

$\bar{v}_{BR} = \bar{v}_B - \bar{v}_R$

$4\hat{j} = \bar{v}_B - 3\hat{i}$

$\bar{v}_B = 3\hat{i} + 4\hat{j} = \bar{v}_F =$ velocity of flag

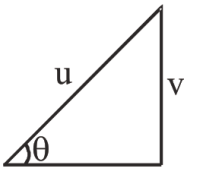
If $\bar{v}_W = k\hat{i}$

$\bar{v}_{WF} = \bar{v}_W - \bar{v}_F$

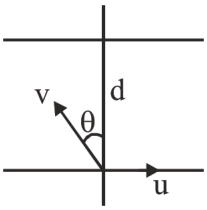
$= K\hat{i} - (3\hat{i} + 4\hat{j})$

If $\bar{v}_{WF}$ points towards south then K = 3 m/s.

28) For min. drift



$$\sin\theta = \frac{v}{u}$$



$$t = \frac{d}{v\cos\theta}$$

$$t = \frac{du}{v\sqrt{u^2 - v^2}}$$

29) $V_1\sin\theta = V_{pilot}$ ...(1)

$V_1\cos\theta = V_2$ ...(2)

Solving equation (1) & (2),

$V_1^2 = V_{pilot}^2 + V_2^2$

$\Rightarrow V_{pilot} = \sqrt{V_1^2 - V_2^2}$

30) $y = \frac{x^2}{(x+1)}$

$\left(\text{apply } y = \frac{u}{v} \rightarrow \frac{dy}{dx} = \frac{u'v - uv'}{v^2}\right)$

$$\frac{dy}{dx} = \frac{(x+1)(2x) - x^2}{(x+1)^2}$$

$$\Rightarrow \frac{dy}{dx} = \frac{x^2 + 2x}{(x+1)^2}$$

Let u_1, u_2, u_3 and u_4 be velocities at time $t = 0, t_1, (t_1 + t_2)$ and $(t_1 + t_2 + t_3)$ respectively and acceleration is a then

$v_1 = \frac{u_1 + u_2}{2}, v_2 = \frac{u_2 + u_3}{2}$ and $v_3 = \frac{u_3 + u_4}{2}$

Also $u_2 = u_1 + at_1, u_3 = u_1 + a(t_1 + t_2)$

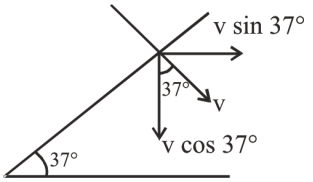
and $u_4 = u_1 + a(t_1 + t_2 + t_3)$

By solving, we get $\frac{v_1 - v_1}{v_2 - v_3} = \frac{(t_1 + t_2)}{(t_2 + t_3)}$.

31)
Correct Ans : (2)

32) $\vec{v} = \frac{d\vec{r}}{dt}$
if $\vec{v} \cdot \vec{r}$ is +ive $|\vec{r}|$ is increases with time $\vec{r} = x\hat{i} + y\hat{j}$
$\vec{v} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j}$
If $\vec{v} \cdot \vec{r}$ is –ve is $|\vec{r}|$ decreases with time & particle is moving towards origin $\vec{v} \cdot \vec{r} < 0$
$x\frac{dx}{dt} + y\frac{dy}{dt} < 0$

33) u_x = vsin 37°
$\Rightarrow \frac{3v}{5} = 15 \Rightarrow v = 25$ m/s



and –vcos 37° = 0 – gt
$\Rightarrow -25 \times \frac{4}{5} = 0 - 10t$
$\Rightarrow$ t = 2 sec

34)

$s_1 = 3 \times 2 = 6$m
$v_2 = 3^2 + 2 \times 10 \times 2 \Rightarrow v = 7$ m/s
$s_2 = 7 \times 3 = 21$ m
total distance = 6 + 2 + 21 = 29 m

35) $\frac{dx}{dt} = v_0 + \lambda x$

$$\int_0^x \frac{dx}{(v_0 + \lambda x)} = \int_0^t dt$$

$$\frac{\ell n[v_0 + \lambda x]_0^x}{\lambda} = t$$

$$\ell n\left(\frac{v_0 + \lambda x}{v_0}\right) = \lambda t$$

$v_0 + \lambda x = v_0 e^{\lambda t}$

$$x = \frac{v_0}{\lambda}\left(e^{\lambda t} - 1\right)$$

$$v = \frac{dx}{dt} = \frac{v_0}{\lambda}\left(e^{\lambda t}\right).\lambda$$

$v = v_0 e^{\lambda t}$

36) $$\frac{\log a}{\log v} = \tan 37^\circ$$

$a = v^{3/4}$

Now integrate acceleration w.r.t. time.

37) $$v = \int a\,dt = \frac{6t^2}{2} + 4t = 3t^2 + 4t$$

Integrate again

$$x = \int v\,dt = \frac{3t^3}{3} + \frac{4t^2}{2} = t^3 + 2t^2$$ At t = 3s, $x = (3)^3 + 2(3)^2 = 45$ m

38) **The assertion is false, and the reason is also false.** If two people jump from the same building, they will initially be in freefall with the same acceleration (due to gravity). Therefore, their relative speed will be zero, and they will not appear to move relative to each other. They would only start moving apart after some time due to minor differences in their initial positions or velocities.

39) Time of flight ∝ vertical component of initial vel.

As $(V_y)_A > (V_y)_B$

$\Rightarrow T_A > T_B$

40)

When the girl moves at $v_{gl} = 5\ \text{ms}^{-1}$, rain appears vertical.

This means the horizontal component of the rain's velocity relative to the ground, v_{rx}, must be equal to the girl's speed.

$v_{rx} = v_{gl} = 5\ \text{ms}^{-1}$.

When the girl moves at $v_{g2} = 15\ \text{ms}^{-1}$, rain appears at 45° to the vertical.

The horizontal component of the rain's velocity relative to the girl is $v_{rx} - v_{g2} = 5 - 15 = -10\ \text{ms}^{-1}$.

Let the vertical component of the rain's velocity be v_{ry}.

The angle $\theta = 45°$ is given by $\tan\theta = \frac{|v_{rx} - v_{g2}|}{|v_{ry}|}$.

$\tan 45° = \frac{|-10|}{|v_{ry}|}$.

$1 = \frac{10}{|v_{ry}|}$.

$|v_{ry}| = 10\ \text{ms}^{-1}$.

The magnitude of the rain's velocity is $|\vec{v}_r| = \sqrt{v_{rx}^2 + v_{ry}^2}$.

$|\vec{v}_r| = \sqrt{(5)^2 + (10)^2}$.

$|\vec{v}_r| = \sqrt{25 + 100}$.

$|\vec{v}_r| = \sqrt{125}$.

$|\vec{v}_r| = \sqrt{25 \times 5} = 5\sqrt{5}\ \text{ms}^{-1}$.

When the girl moves at $v_{g1} = 5\ \text{ms}^{-1}$, rain appears vertical.

This means the horizontal component of the rain's velocity relative to the ground, v_{rx}, must be equal to the girl's speed.

$v_{rx} = v_{g1} = 5\ \text{ms}^{-1}$.

41)

Since the trajectories are identical and the angle is the same, the ratio $\frac{g}{u^2}$ must be equal for both cases.

$\frac{g_E}{u_E^2} = \frac{g_P}{u_P^2}$

We have $g_E = 9.8$, $u_E = 5$, and $u_P = 3$.

$\frac{9.8}{5^2} = \frac{g_P}{3^2}$

The acceleration due to gravity on the planet is 3.528 ms^{-2}

42)

Option (1) is correct

43) Relative velocity does not depend upon observer

$|\vec{v}_A - \vec{v}_C| = |(\vec{v}_A - \vec{v}_B) - (\vec{v}_C - \vec{v}_B)|$

$= 3 + 4$

$= 7\ m/s$

(D)

44)

Option (1) is correct

$$500 \cos\theta = 250$$

$$\Rightarrow \cos\theta = \frac{1}{2}$$

45) $\Rightarrow \theta = 60^\circ$

CHEMISTRY

46)

$S_2O_3^{-2} \longrightarrow H_2S + SO_4^{-2}$

Ox. No. = +2 −2 +6

8 = n factor

$$E = \frac{M}{8}$$

47) $KMnO_4$ = Mohr salt

$$\frac{M_1V_1}{n_1} = \frac{M_2V_2}{n_2} \; ; \; \frac{0.1 \times 10}{1} = \frac{M_2V_2}{5} \; ; \; M_2V_2 = 5 .$$

48) When sulphur dioxide is react with H_2S here SO_2 act as an oxidising agent and H_2S act as reducing agent.

49)

$NH_4^+ \longrightarrow N_2$

(Ox. State = − 3) (Ox. state = Zero)

50) For compound $X_3(YZ_4)_2$, Sum of Ox. states is zero.

51) $1.5 \times 10^{-3} \times (10-2x) = 0.03 \times 5 \times \frac{40}{1000}$

$x = 3$

52) $S_2O_3 \rightarrow S_4O_6^{-2}$ nf = 1

$cu^{+1} \rightarrow cu^{+1}$ nf = 1

53) $2e^- + H_3AsO_4 \longrightarrow H_3AsO_3$

$+5 \longrightarrow +3$

$2I^- \longrightarrow I_2 + 2e^-$

Mass of $I_2 = \frac{1.5 \times 10^{22}}{2 \times N} \times 254$

$\Rightarrow \frac{1.5 \times 10^{22}}{2 \times 6 \times 6^{23}} \times 254 = 3.2$ gm

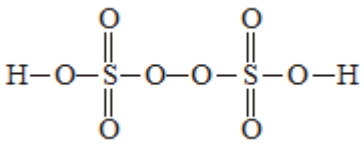
54)

In both molecules, hydrogen (H) has an oxidation number of +1 and nitrogen (N) has an oxidation number of -3. The overall charge of the molecule is zero. Therefore, to balance the charges, the oxidation number of carbon (C) must be +2 in both cases:

A. For HCN: (+1) + (C) + (-3) = 0 => (C) = +2
B. For HNC: (+1) + (C) + (-3) = 0 => (C) = +2

55) $5H_2O_2 + 2BrO_2 + 2OH^- \longrightarrow 2Br^- + 5O_2 + 6H_2O$;

56) $\overset{1.5\ -2}{P_4S_3} \longrightarrow \underset{14e^-}{\overset{+5}{2P_2O_5}} + \underset{18e^-}{\overset{+4}{3SO_2}}$

57) 

According to structure,
Oxidation state of 2"O" atom = –1
(due to peroxide)
Oxidation state of 6"O" atom = –2
 Ratio of atoms = 2 : 6 = 1 : 3

58) $Na_2\overset{(+2)}{S_2}O_3 \rightarrow Na_2\overset{(+2.5)}{S_4}O_6$

n-factor (e– transfer per mole)

= (+ 2 to + 2.5) × 2 "S atom = 1

$$E_{Na_2S_2O_3} = \frac{\text{Molar mass}}{\text{n} - \text{factor}} = \frac{158}{1}$$

$$\overset{(+7)}{KMnO_4} \xrightarrow[\text{medium}]{\text{basic}} K_2\overset{+6}{MnO_4}$$

n-factor = (+ 7 to + 6) × 1 "Mn" atom

$$E_{KMnO_4} = \frac{\text{Molar mass}}{1} = \frac{158}{1}$$

59) $H_3PO_2 \rightarrow PH_3 + H_3PO_3$

$$\frac{4 \times 2}{4+2} = \frac{8}{6} = \frac{4}{3}$$

Eq. Mass = $\frac{M}{4/3}$ = 3M/4

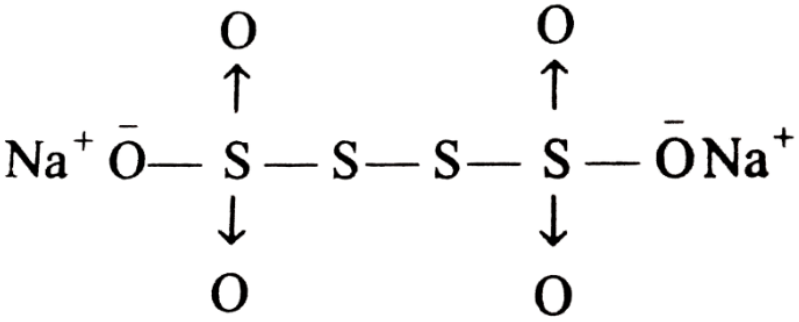
60) Let the oxidation number of oxygen is x

+ 3 + 2x + 0 × 4 + 0 – 2 = 0

2x + 1 = 0 ⇒ 2x = –1

$\left[x = -\frac{1}{2}\right]$

Thus, oxygen will in superoxo form.



61)

diff = 5 – 0 = 5

62) $BrO_3 + Br^- - \rightarrow Br_2^O$

nf = 5

E = M/5

63)

Equivalents = Mole × (n-factor) = Faraday

$MnO_4^- \rightarrow Mn^{+2}$; 1 mole × 5 = 5F = 482500 C

$Cr_2O_7^{-2} \rightarrow Cr^{+3}$; 1 mole × 6 = 6F = 579000C

$Sn^{+4} \rightarrow Sn^{+2}$; 1 mole × 2 = 2F = 193000 C

$Al^{+3} \rightarrow Al$; 1 mole × 3 = 3F = 289500 C

64) Caro's acid = H_2SO_5
= Peroxomono-sulphuric acid
Marshell's acid = $H_2S_2O_8$
= Peroxodi-sulphuric acid
The oxidation state in both the acids is +6 each because oxidation state cannot be greater than the number of valuence electrons.

65)

In reaction
I, II, III, No. of replaced
Hydrogens are 1,2,3,
respectively, so valency factors are 1,2,3 respectively

66) $P_2H_4 \rightarrow PH_3 + P_HH_2$
nf = $\frac{2 \times 3}{2+3} = \frac{6}{5}$
Equivalent Mass = $\frac{5m}{6}$

67) $Fe_{0.9}O \rightarrow Fe_2O_3$
nf = 0.72
$E = \frac{M}{0.7}$

68)

In a conjugate pair of reductant and oxidant, the oxidant has a higher oxidation number. This is because the oxidant is the species that accepts electrons, causing its oxidation number to decrease (reduction), while the reductant loses electrons and increases its oxidation number

69)

Correct Opiton is (2)

70) $O_2^{-2} \xrightarrow{H+} O_{-2}^{-2}$
Per'O' = –1

71) $FeC_2O_4 + Cr2O_7^{-2} \rightarrow Fe + CO_2 + Cr$
nf = 3 nf = 6
(e) FeC_2O_4 = (e) $Cr_2O_7^{-2}$
1×3 = x × 6 = x = $\frac{1}{2}$

72)

Disproportionation where oxidation and reduction occurs both.

73)

The balanced equation is
$2C_2H_6 + 7O_2$ â†’ $4CO_2 + 6H_2O$.
Ratio of the coefficients of CO_2 and H_2O is 4 : 6 or 2 : 3

74) Eq. of metal = Eq. of oxygen

$$\text{or} \quad \frac{100}{E} = \frac{24}{8}$$

$$\therefore \quad E = 33.3$$

$$P_2H_4 \xrightarrow{-10e^-} \underset{\text{[Compound Y]}}{(P_2H_4)^{+10}}$$

75) in compound Y;

$2x + 4 = +10$

$2x = +6 \Rightarrow [x = +3]$

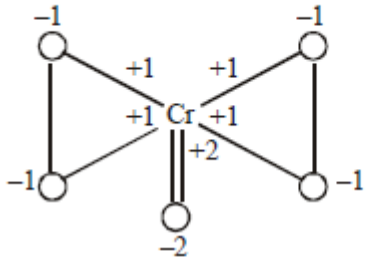
76)

element	%	mole	mole ratio
X_e	53.5	$\frac{53.5}{131} = 0.4$	$\frac{0.4}{0.4} = 1$
F	46.5	$\frac{46.5}{19} = 2.4$	$\frac{2.4}{0.4} = 6$

Hence compound may be XeF_6.

$Sn^{+2} \longrightarrow Sn^{+4}$ (oxidation)

77)



(CrO_5)

$\Rightarrow$ In CrO_5, 4 oxygln atom show –1 oxidation state and oxidation number of Cr = +6.

In CrO_4^{-2}; Oxidation number of C_r = +6

and oxidation number of oxygen = –2

78) **Question Explanation:**
Choose correct option.

Concept:
This question is based on Equivalent concept.

Solution:
$K\overset{+7}{Mn}O_4 \rightarrow Mn^{+2}$
eq. of $KMnO_4$ = M × V × n.f = 0.1 × v × 5
= 0.5 V
(1) $C_2O_4^{-2} \rightarrow CO_2$
eq. of $C_2O_4^{-2}$ = 0.25 × 2 × V = 0.5 V
(2) $Fe^{+2} \rightarrow Fe^{+3}$
eq. of Fe^{+2} = 0.8 × 1 × V = 0.8 V
(3) $FeC_2O_4 \rightarrow Fe^{+3} + CO_2$
n.f.: 3
eq. of FeC_2O_4 = 3 × 0.1 × V = 0.3 V
(4) $Cr_2O_7^{-2} \rightarrow Cr^{+3}$
n.f. = 6
eq. of $Cr_2O_7^{-2}$ = 0.6 × 6 × V = 3.6 V

Final Answer:
The correct option is (1).

79) $(N_1 \times V_1)$metal = $(N_2 \times V_2)KMnO_4$
0.1 M × (n - factor) × 25 ml
= 0.04 M × 5 × 25 ml
n-factor of metal = 2
only true for $Z^{+2} \rightarrow Z^{+4}$

80) M_4O_5
$M_x(+1)\ M_y(+3)\ O_5$
x + y = 4

x + 3y = 10	x + 3y = 10
−2y = −6	−2y = −6

y = 3 ; x = 1

81)

$N_2H_4 + BrO_3^- \rightarrow Br^- + N_2^O$
(nf = 4) (nf = 6)
(e)N_2H_4 = (e)BrO_3^-
$x \times 4 = \frac{2}{3} \times 6$
x = 1

82)

Option (1) is correct

83) $\text{eq}\,(\overset{+3}{Fe}_2(\overset{+3}{C}_2O_4)_3 + \overset{+2}{Fe}\,\overset{+3}{C}_2O_4) \equiv (K\overset{+7}{M}nO_4)\text{eq.}$

n.f. (6) (1 + 2) = 3 (5)

equimolar i.e. $\frac{1}{2}$ Mole each

$\frac{1}{2} \times 6 + \frac{1}{2} \times 3 = x \times 5$

x = 0.9

84)

Option (2) is correct

$$\overset{+2}{CuSO_4} + 2KI \rightarrow K_2SO_4 + \overset{+2}{CuI_2} \;;\; 2\overset{+2}{CuI_2} \rightarrow \overset{+1}{CuI_2} + I_2$$

$$I_2 + 2Na_2S_2O_3 \rightarrow 2NaI + Na_2S_4O_6$$

85) Eq. wt. Of $CuSO_4.5H_2O$ = Mol. wt. = 250

100 ml of 0.1 N hypo 100 ml of 0.1 N

$$CuSO_4.5H_2O = \frac{250 \times 0.1 \times 100}{100} = 2.5 \text{ gm}$$

86) $Cr_2O_7^{2-}$ (n = 6) + $6Fe^{2+}$ (n = 1) (Mohr's salt)
+ 14 $H^+ \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$
Equivalent of Fe^{2+} = moles of Mohr's salt
= equivalent of $K_2Cr_2O_7$
$= 500 \times 10^{-3} \times 6 \times 1 = 3.0$
Hence, mole percent of Mohr's salt
$= \frac{3}{4} \times 100 = 75$

87) meq. $FeSO_4(NH_4)_2SO_4.6H_2O$ = meq of $KMnO_4$
(n = 1)

$\frac{W}{392} \times 1 \times 1000 = 0.1 \times 50$; W = 1.96 g

Hence, % purity of Mohr's salt

$= \frac{1.96}{2.5} \times 100 = 78.4\%$

88) For complete reaction;

geq of $K_2Cr_2O_7$ = geq of Xth

or mole of $K_2Cr_2O_7 \times (\text{valency h factor})_{K_2Cr_2O_7}$

$= \text{moles of } X^{th} \times (\text{valency factor})_{X^{th}}$

$6 \times 10^{-3} \times 6 = (5-n) \times 9 \times 10^{-3}$

[n =1]

89) $Fe_2(C_2O_4)_3 \rightarrow 2Fe^{+++} + 6CO_2$ [n.factor = 6]

$Br_2 \rightarrow Br^- + BrO_3^-$ [n – factor = 5/3]

$SnCl_2 + HCl_2 + O_3 \rightarrow \underset{+SnCl_4}{H_2O}$ $\left[n - \underset{\text{of } O_3}{\text{factors}} = 6\right]$

$S_2O_3^{--} \rightarrow SO_4^{--}$ (n – factors = 8)

90) $\overset{-n}{ABD} + Cr_2^{+6}O_7^{2-}$

$2Cr^{3+} + Ay$

Total oxidation = Total Reduction

$(3.36 \times 10^{-3})\ (y - (-n)) = (1.68 \times 10^{-3}) \times 6$

$2 \times (y + n) = 6$

$y = 3 - n$

BIOLOGY

91) NCERT Pg. # 165,166

92) NCERT Pg. # 163

93)

NCERT Pg. No. 126

94)

SOLUTION —

Most of the histone production occurs during the S phase of the cell cycle, which is when DNA replication takes place. As the DNA replicates, histones are synthesized to package the newly synthesized DNA into chromatin.

The correct answer is: OPTION(3) S phase

95) NCERT XI Pg. # 123

96)

Generated by Allie

Problem Statement: The question asks how many of the given four statements about meiosis are correct. Each statement describes a specific feature or event during the process of meiosis.

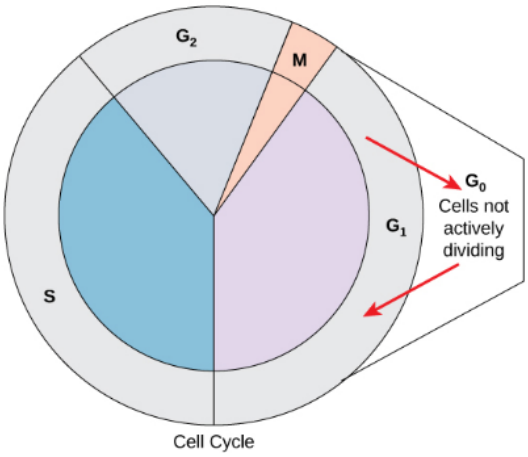
Underlying Concept: Meiosis is a special type of cell division that reduces the chromosome number by half, producing four haploid cells from one diploid cell. It involves DNA replication, pairing of homologous chromosomes, recombination, and two rounds of division (meiosis I and II). This process is essential for sexual reproduction and genetic diversity.

Tips and Tricks: Remember the key distinctions: Meiosis I separates homologous chromosomes, and meiosis II separates sister chromatids. Also, recall that meiosis results in haploid gametes or spores.

Common Mistakes: Confusing meiosis I with meiosis II, believing that DNA replicates before meiosis II, or assuming meiosis produces diploid cells instead of haploid.

97)

The most variable phase of the cell cycle is G1 (Gap 1). This is the phase where cells grow and synthesize proteins and organelles, preparing for DNA replication, and its duration can fluctuate significantly based on various factors, according to biology resources.



98)

To solve the question "Ends of chromosomes are attached with nuclear envelope at attachment plate in," we need to analyze the stages of prophase I of meiosis and identify which stage corresponds to the attachment of chromosome ends to the nuclear envelope.

1. **Understanding the Stages of Meiosis**:
- Meiosis consists of two rounds of cell division: meiosis I and meiosis II. Prophase I is further divided into several stages: leptotene, zygotene, pachytene, diplotene, and diakinesis.

2. **Identifying the Stages**:
- **Leptotene**: In this initial stage, chromosomes start to condense and become visible. The ends of the chromosomes attach to the nuclear envelope at the attachment plate.

- **Zygotene**: This stage follows leptotene, where homologous chromosomes begin to pair up (synapsis). However, the attachment to the nuclear envelope is not the primary characteristic of this stage.
- **Pachytene**: At this stage, crossing over occurs between homologous chromosomes, but the attachment to the nuclear envelope is not emphasized.
- **Diplotene**: In this stage, synapsis ends, and the chromosomes start to separate, but they remain attached at crossover points.

3. **Conclusion**:
- The correct answer is **leptotene**, as it is the stage where the ends of chromosomes are attached to the nuclear envelope at the attachment plate.

Final Answer:
The ends of chromosomes are attached with the nuclear envelope at the attachment plate in **leptotene**

99)

Answer: 15

A. During zygotene, which is a stage of prophase I of meiosis, homologous chromosomes pair up to form bivalents. Each pair of homologous chromosomes forms one bivalent.
B. Since there are 15 pairs of chromosomes in a diploid cell with 30 chromosomes, the number of bivalents formed during zygotene will be 15.

Thus, the number of bivalents is 15.

100)

NCERT XI, Page # 162,163,164,165

101)

SOLUTION —
In a cell with 2C DNA in metaphase, the DNA amount in anaphase (x) is 2C, in G1 phase (y) is C, and after M-phase (z) is C.

The correct answer is: OPTION(3) (x)-2c, (y) -c, (z)-c

102)

Explain Question : This question is asking to find the incorrect match where one or both events do not occur in the stated phase.

Concept : This question is based on Cell cycle phases: G_1, S, G_2, M phase

Explanation : Anaphase

(i) Splitting of centromere – Correct

Happens during anaphase of mitosis/meiosis II.

(ii) Movement of centrioles toward opposite poles – Incorrect

This occurs during prophase, not anaphase. During prophase, centrioles move to opposite poles to set up the spindle apparatus.

Final answer : 2

103)

NCERT Line

104)

After meiosis, a spore will have 5 picograms (pg) of DNA. In the G1 stage of the spore mother cell, the DNA content is 10 pg. After DNA replication in the S phase, it doubles to 20 pg before meiosis I. Meiosis I reduces the DNA content to 10 pg in each of the two daughter cells. Meiosis II further divides the DNA content in each daughter cell, resulting in **5 pg** of DNA in each of the four spores.

105)

Option (2) is correct

106)

Correct answer is Anaphase-II, it means the figure must show the following features:

A. Separation of sister chromatids: In Anaphase-II of meiosis, the sister chromatids are pulled apart to opposite poles of the cell by the spindle fibers.
B. This stage occurs after the second metaphase, where the chromatids align individually at the equator.

107)

Ncert Page 121.

108)

Option (1) is correct

109)

NCERT Line

110)

SOLUTION—

Meiosis involves two rounds of cell division, meiosis I and meiosis II, but only one round of DNA replication, which occurs before meiosis I.

The correct answer is: OPTION (4) A single DNA replication and two divisions of cytoplasm

111)

SOLUTION—

Synapsis occurs between two homologous chromosomes. This pairing is a crucial step in meiosis, specifically during prophase I, and allows for genetic material exchange through crossing over.

The correct answer is: OPTION (2) Two homologous chromosomes

112)

SOLUTION—

In late prophase, also sometimes called prometaphase, the nuclear membrane and nucleolus begin to disappear.
The correct answer is: OPTION (4) late prophase

113) **Solution:**
During zygotene, homologous chromosomes pair up in a process called synapsis, forming the bivalent or tetrad structure.

The correct answer is option (3) zygotene.

114)

In a cell with 16 tetrads in the G1 phase, there will be 64 chromatids in metaphase I and 32 chromatids in metaphase II.

115) NCERT-XI, Pg. # 122,126

116) NCERT Page. No. 168,169

117)

All five statements (A, B, C, D, and E) are correct with respect to prophase

118)

1. **Understanding the Term "Tetrad":**
- The term "tetrad" is derived from the Greek word "tetra," which means four. In the context of cell division, particularly during meiosis, a tetrad refers to a structure that consists of four

chromatids.

2. **Identifying Chromatids in Meiosis:**
- During meiosis, specifically in the prophase stage, homologous chromosomes pair up. Each chromosome consists of two sister chromatids. Therefore, when two homologous chromosomes come together, they form a tetrad, which contains a total of four chromatids.

3. **Phases of Meiosis:**
- The formation of tetrads occurs during the zygotene stage of prophase I of meiosis. This is where homologous chromosomes align closely together and begin to pair up.

4. **Characteristics of Tetrads:**
- A tetrad consists of two homologous chromosomes, each made up of two sister chromatids. Thus, the tetrad is made up of four chromatids in total.

5. **Conclusion:**
- Therefore, a tetrad is made of **four chromatids** that are formed from the pairing of **two homologous chromosomes** during the zygotene phase of meiosis.

Final Answer:
A tetrad is made of **four chromatids** (two homologous chromosomes, each with two sister chromatids).

119) NCERT Pg : 125, 126 (new)

120)

Synapsis occur in zygotene.

121)

SOLUTION —

A - G0 phase:
Cells like heart and nerve cells, which are non-dividing, are in the G0 phase, also known as the quiescent phase.

B - Nucleus:
During the S phase in animal cells, DNA replication occurs in the nucleus.

C - Cytoplasm:
During the S phase in animal cells, centriole duplication occurs in the cytoplasm.

The correct answer is: OPTION(1)

A-Go

B - nucleus,

C- cytoplasm

122)

SOLUTION —
Many cells in adult animals exit the G1 phase (A) to enter an inactive stage called the quiescent stage (B) or G0 phase, where they remain metabolically active (C) but do not divide (D).

The correct answer is: OPTION(4) A = G_1 phase, B = Quiescent stage, C = active, D = not divide

123)

Option (4) is correct

124) **Explanation**

A. In the pachytene stage of meiosis-I, each bivalent chromosome consists of two homologous chromosomes, each with two chromatids. Therefore, a bivalent will have a total of four chromatids.

These four chromatids represent the two homologous chromosomes, each with two sister chromatids, making the total number of chromatids four.

Correct Answer: 2. Four

125)

1. **Understand the term "Amitosis":**
- Amitosis is a type of cell division that is characterized by the direct division of the nucleus without the formation of chromosomes. This process results in the formation of two daughter nuclei.

2. **Identify the characteristics of Amitosis:**
- Unlike mitosis or meiosis, amitosis does not involve the complex stages of chromosomal alignment and separation. Instead, the nucleus simply splits into two parts, each becoming a daughter nucleus.

3. **Determine where Amitosis occurs:**
- Amitosis is primarily observed in prokaryotic cells, such as bacteria. Prokaryotic cells lack a defined nucleus and do not undergo the typical mitotic processes seen in eukaryotic cells.

4. **Conclude the answer:**
- Based on the characteristics and the occurrence of amitosis, the correct answer to the question "Amitosis usually occurs in" is prokaryotic cells.

Final Answer:
Amitosis usually occurs in prokaryotic cells

126)

In anaphase, the centromere splits and chromosomes move apart towards two opposite poles due to shortening of spindle fibers

127)

NCERT Line

128) In some organisms, division of the nucleus occurs but division of the cytoplasm does not occur. As a result, a large number of nuclei are present in the cytoplasm. This leads to the formation of syncytium; that is, a cell containing a large number of nuclei.

129) Cyclins are proteins that activate protein kinases to regulate the eukaryotic cell cycle.

130)

NCERT Line

131) NCERT Pg. # 167, 168

132)

NCERT Pg.#168

133) NCERT XI P. No. # 169, 170

134)

NCERT Line

135)

Anaphase Promoting Complex (APC) is a protein necessary for separation of daughter chromosomes during anaphase. If APC is defective then the chromosomes will fail to segregate during anaphase

136)

Question Explanation : Cell composition-abundance

Concept : This question is based on Cell composition analysis.

Solution : Water > Protein > Nucleic acid > Carbohydrate > Ions.

Final Answer : option (3)

137)

Explanation:
Polyunsaturated fatty acids have multiple double bonds in their carbon chain.

Linoleic, linolenic, and arachidonic acids all have multiple double bonds.

The other options either have one double bond (oleic) or no double bonds (palmitic and stearic).

Correct Answer: Option 4 (Linoleic, linolenic, arachidonic)

138) NCERT-XI Pg#106

139)

Concept:

A. The living state is described as non-equilibrium because living systems are constantly undergoing reactions and processes that maintain this dynamic, non-steady state.

Explanation:
Both statements are correct:

A. Statement 1 is true because living organisms require constant energy input to maintain their non-equilibrium state.

B. Statement 2 is true because the living state is inherently a non-equilibrium state, where various biological processes prevent equilibrium.

140)

Nucleic acids that behave like enzymes are called ribozymes. Ribozymes are molecules of ribonucleic acid (RNA) that exhibit catalytic activity, meaning they can speed up chemical reactions, much like proteins that act as enzymes.

141)

Question Explanation : Correct statement

Concept : This question is based on Lipids

Solution :

A. Lipids- water insoluble
    A. R-Group $-CH_3$, $-CH_2$, $-C_2H_5$

B. Palmitic Acid - 16C
    A. A, B, C are correct

Final Answer : option (1)

142)

Concept:

A. Sweetness: The relative taste intensity of sugars.

B. Fructose: A monosaccharide (fruit sugar) found in fruits and honey.

Explanation:

A. Fructose is known to be the sweetest naturally occurring sugar.

Final Answer:

A. **The correct answer is Fructose (1).**

143) **Explanation :**
In DNA, N-Glycosidic bonds link the sugar (deoxyribose) to the nitrogenous bases (adenine, thymine, cytosine, and guanine).

144)

Concept

A. **Essential amino acids:** Amino acids that cannot be synthesized by the human body and must be obtained from the diet.

Explanation

A. Tryptophan is one of the nine essential amino acids that humans cannot synthesize.

Final Answer : The correct answer is 3. Tryptophan.

145)

NCERT XI Pg # 107

146)

NCERT XI Pg # 105

147)

NCERT-XI, Pg # 147

148)

Pg. No. 146_XI NCERT

149) NCERT Pg. No. # 149

150) NCERT Pg # 145

151)

NCERT XI Pg # 106

152)

NCERT XI Pg.# 146

153)

Concept:

A. Cofactors are molecules or ions required by enzymes for catalytic activity.

B. Coenzymes are a subset of cofactors that are organic molecules, often derived from vitamins.

Explanation:

A. Statement 1: This is incorrect. Not every coenzyme is a cofactor, and not every cofactor is a coenzyme. Coenzymes are a type of cofactor, but not all cofactors are coenzymes. Some cofactors are inorganic ions (e.g., Mg^{2+}, Zn^{2+}).

154)

Option 4: A and C

(A) This graph is for an exothermic process: This statement IS incorrect. It is an endothermic process..

(C) Graph 'a' shows the presence of enzymes: This statement IS incorrect.

155)

NCERT XI Pg # 113

156) **Concept; Hydrolases:** Catalyze hydrolysis reactions[1] (breaking bonds with water).
Explanation; Hydrolases are the class of enzymes that catalyze hydrolysis reactions.

157)

NCERT XIth, Page 143

158)

NCERT XI Pg # 110

159) NCERT XI, Page # 145

160) NCERT-XI
Hint Page -142, Trichloroacetic acid is used as a medium to grind plant or animal tissue.

161)

NCERT XI Pg # 110

162) **Concept :**
• Polysaccharide : A complex carbohydrate made up of many monosaccharide units linked together.

Explanation :
• In a polysaccharide chain, only one end will have a free anomeric carbon. This end is the reducing end.
• The other end of the chain will have its anomeric carbon involved in a glycosidic bond, making it a non-reducing end.
• Therefore, the right end is the reducing end, and the left end is the non-reducing end.

163)

Chargaff Rule

164)

NCERT XI Pg # 106

165)

NCERT XI Pg # 111

166)

NCERT XI Pg # 111

167)

NCERT XI Pg # 117

168)

NCERT XI Pg # 115

169)

NCERT XI Pg # 105

170)

NCERT XI Pg # 107

171)

NCERT XI Pg # 106

172)

The Correct Answer IS 3.
The structure of keratin in mammalian hairs is α-helical, unlike silk fibroin.
Keratin in Mammalian Hairs
Keratin in mammalian hairs is primarily composed of **α-helices.**
Silk Fibroin
Silk fibroin, on the other hand, is primarily composed of **β-pleated sheets**

173)

NCERT XI Pg # 107

174)

NCERT XI Pg # 117

175)

Option 2: A - Lecithin, B - Triglyceride is the correct answer.
Lipid A matches the structure of lecithin (a type of phospholipid).
Lipid B matches the structure of a **triglyceride.**

176)

NCERT XI Pg # 107

177)

NCERT XI Pg # 112

178)

NCERT XI Pg # 178

179)

NCERT XI Pg # 106

180)

Question Explanation : Correct statement

Concept : This question is based on Polysaccharide - Carbohydrates

Solution :
Arthropods- exoskeleton → Chitin
→ Starch helical structure
→ N-acetyl-D-Galactosamine is modified sugar
These are correct statement.
a, b, c

Final Answer : option (3)